

Thesis for the degree of Master of Science

Design and Implementation of a Multimodal
Adaptive Guidance System for Procedural Task
Assistance in Mixed Reality

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Design and Implementation of a Multimodal Adaptive Guidance System for Procedural Task Assistance in Mixed Reality

A Thesis submitted to the graduate school of Pusan National University in partial fulfillment of the requirements for the degree of Master of Science under the direction of **Myungho Lee**

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Abstract

Mixed reality (MR) guidance systems predominantly rely on static instructions, failing to account for the cognitive and behavioral variability of real-world users. This thesis presents a multimodal adaptive guidance system for procedural laboratory training in mixed reality, implemented on the Microsoft HoloLens 2 platform. The proposed system integrates structured task representation, real-time semantic step tracking, multimodal user state estimation, and LLM-based adaptive guidance into a unified closed-loop architecture. Task progress is modeled as a directed acyclic graph (DAG) generated from natural-language protocols using a large language model (LLM), enabling automatic and flexible task structuring, and is tracked in real time through a multimodal action recognition pipeline leveraging sensor data derived from gaze, hand, and head interactions. These multimodal signals are used to estimate the user's procedural difficulty state, enabling the guidance module to dynamically adapt feedback in terms of granularity, tone, and content. Evaluation of the guidance generation component shows that incorporating real-time user state signals alongside task context improves response relevance and reduces hallucination compared to context-only baselines. Experimental results demonstrate robust step tracking performance and low-latency system responsiveness, supporting real-time deployment in immersive laboratory settings. Overall, this work establishes a practical framework for user state-aware adaptive MR guidance, advancing beyond one-size-fits-all instruction toward personalized learning support.

Keywords: Mixed Reality, Large Language Models, Adaptive Learning, Multimodal Data, User State Estimation, AI-Driven Personalization

Chapter 1

Introduction

1.1 Problem Statement

Procedural training and task guidance play a central role in domains where users must perform structured physical actions in the correct order while interacting with real tools and objects. Such tasks are common in technical maintenance, laboratory work, medical assistance, industrial assembly and everyday skill acquisition, where mistakes can reduce efficiency, compromise safety, or prevent successful task completion. Conventional instructional materials, including paper manuals, videos or static digital checklists, provide useful reference information, but they do not actively interpret user behavior during execution or adapt guidance to the evolving state of the task.

Mixed Reality (MR) provides a promising foundation for procedural assistance because it enables digital guidance to be spatially aligned with the physical environment while preserving direct interaction with real objects. Head-mounted MR devices such as Microsoft HoloLens 2 can capture multimodal signals including gaze, hand motion, head pose and interaction events, while simultaneously presenting contextual visual and auditory instructions. This makes MR especially suitable for embodied learning and step-by-step task support. However, moving from static MR instructions to genuinely adaptive procedural guidance remains a challenging systems problem.

Existing work on procedural AR/MR guidance suggests that static instructions can be insufficient for complex real-world tasks, especially when users deviate from the expected sequence, repeat actions, or require additional support. Building on this observation, this thesis treats procedural assistance in MR not only as the presentation of visual overlays, but as a real-time interactive process in which the system interprets user actions, infers observed behavior within the current procedural context and adapts guidance when signs of difficulty or deviation are detected.

This thesis addresses the design of a real-time adaptive MR guidance system for procedural tasks involving real physical objects. The central goal is to support users during task execution by continuously combining low-level multimodal sensing, semantic task-state reasoning and adaptive guidance generation. Rather than treating procedural assistance as a static sequence of instructions, this work frames it as a closed-loop interaction problem in which the system observes user behavior, interprets its procedural meaning, estimates task-related difficulty, and delivers contextual support during runtime.

1.2 Research Gap

Despite substantial progress in Extended Reality (XR), multimodal sensing and AI-assisted interfaces, existing systems still leave a significant gap in supporting real-time adaptive procedural guidance for tasks involving real physical objects.

First, many MR guidance systems focus on presenting step-by-step instructions or spatial cues, but do not maintain a rich semantic representation of procedural state during execution. As a result, they often present guidance at the interface level, yet provide limited reasoning about whether an observed action is expected, premature, repeated or recoverable within the current procedure.

Second, a large body of work on action recognition has been developed in video understanding and egocentric perception. However, these methods are often optimized for offline recognition benchmarks or general activity labels rather than real-time procedural assistance on head-mounted MR devices. In practical MR guidance scenarios, the key question is not only what motion occurred, but what that action means relative to the current task graph, step dependencies and possible recovery paths. This motivates a shift from isolated action labeling towards semantic procedural tracking.

Third, adaptive XR/AR systems frequently discuss personalization in terms of cognitive load, emotion or engagement, yet reliable real-time estimation of such latent user states remains challenging in realistic MR settings. Many approaches depend on physiological sensing, controlled laboratory conditions, or datasets that are difficult to transfer to real-world procedural tasks. For an operational MR guidance system, a more robust formulation is to estimate observable procedural difficulty from multimodal behavioral signals and use it as an adaptive cue.

Finally, recent work has demonstrated the potential of Large Language Models (LLMs) and multimodal models for interactive assistance, instruction generation and contextual reasoning.

Although recent XR/AR assistance systems increasingly explore proactive guidance, object-

aware interaction, and LLM-based procedural support, the specific gap addressed in this thesis is the integration of semantic procedural task tracking with behaviorally grounded procedural difficulty estimation, so that guidance can adapt not only to the current procedural step, but also to the user’s observed difficulty during task execution. However, there remains limited evidence of end-to-end systems that integrate LLM-based procedural representation, real-object semantic tracking, behavioral user-state estimation and real-time adaptive guidance in a closed loop on an MR platform. This thesis addresses this gap by proposing and evaluating a system that unifies these components within a single runtime architecture.

1.3 Research Questions

This thesis is guided by the following research questions:

1. How can multimodal interaction signals in mixed reality be interpreted semantically to support real-time procedural step tracking and outcome classification?
2. Can temporal multimodal behavioral signals be used to estimate user difficulty states that are useful for adaptive procedural guidance in MR environments?
3. How effectively can adaptive guidance be generated from procedural context, semantic outcomes, and estimated procedural difficulty during real-world MR task execution?
4. Does a closed-loop semantic adaptive MR guidance framework improve procedural task support compared with a non-adaptive baseline system?

These research questions reflect the central goal of the thesis: enabling adaptive procedural assistance in mixed reality through semantic task understanding, multimodal behavioral interpretation, and context-aware guidance generation.

1.4 Thesis Contributions

The main contributions of this thesis are as follows:

1. **A closed-loop semantic adaptive guidance framework for procedural mixed reality tasks.** This thesis proposes a unified mixed reality framework in which multimodal user interactions are interpreted relative to procedural context, semantic task state and inferred

user difficulty in order to support real-time adaptive guidance during physical task execution. Unlike conventional step-based XR assistance systems, the proposed framework treats procedural guidance as a continuous closed-loop process of sensing, semantic interpretation, runtime reasoning and adaptive intervention.

2. **A semantic procedural tracking approach for real-object MR interaction.** The thesis introduces a real-time procedural tracking approach that combines multimodal interaction signals, object-centric sensing and procedural state management to interpret user actions semantically rather than as isolated low-level events. The proposed approach supports runtime procedural reasoning over task progression, action validity and recovery behavior within dependency-aware task execution.
3. **A multimodal temporal user-state estimation method for adaptive procedural assistance.** The thesis presents a temporal modeling approach for estimating user difficulty states from gaze, hand, and head-motion dynamics captured during procedural task execution. Rather than attempting to directly infer latent cognitive or affective states, the proposed method operationalizes behaviorally expressed procedural difficulty as an adaptive signal for real-time intervention and support.
4. **A context-aware adaptive guidance mechanism for mixed reality procedural assistance.** The proposed system dynamically generates guidance based on procedural state, semantic outcomes and inferred user difficulty to provide contextual verbal and visual assistance during task execution. This adaptive mechanism enables intervention-aware support that responds to procedural mistakes, uncertainty and difficulty progression in real time.

In addition to these methodological contributions, the thesis presents an integrated runtime implementation on Microsoft HoloLens 2 and evaluates the proposed framework through a comparative user study and runtime performance analysis.

1.5 Overview of the Proposed System

The proposed system consists of four tightly connected components. The first component constructs a procedural task graph from a natural-language description of the target task. This graph acts as an executable semantic representation of the procedure, capturing steps, dependencies, and

allowable transitions. The second component performs semantic step tracking during task execution. Using object detection, hand-object interaction cues, gaze, and MR interaction events, the system identifies user actions and interprets their meaning relative to the procedural graph.

The third component estimates user state in terms of observable procedural difficulty. Temporal multimodal signals are aggregated and processed by a recurrent model to distinguish comparatively stable interaction periods from behavior associated with struggle or increased task difficulty. This signal is not treated as a direct measure of cognition, but as an operational indicator that helps determine when additional support may be needed.

The fourth component generates adaptive guidance. Based on the current procedural state, semantic outcome, and estimated difficulty level, the system constructs a contextual prompt for a local or backend language model, which returns structured guidance for both verbal and visual feedback. In this way, the system forms a closed loop from sensing to interpretation to adaptation. The resulting architecture aims to reduce user confusion, improve support during errors or difficult moments, and provide context-sensitive assistance without requiring fully manual authoring of all instructional behavior.

1.6 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews the background and related work relevant to adaptive XR systems, multimodal sensing, action and task tracking, user-state estimation, and language-model-based guidance. Chapter 3 presents the methodology of the proposed system, including task graph construction, semantic step tracking, observable procedural difficulty modeling, and adaptive guidance generation. Chapter 4 describes the system implementation on HoloLens 2, focusing on the hardware and software stack, runtime architecture, and latency-aware engineering considerations. Chapter 5 reports the evaluation methodology and results, including the comparative user study, quantitative findings, and post-hoc analysis of adaptive guidance behavior. Chapter 6 discusses the interpretation of results, limitations, threats to validity, and future work. Finally, Chapter 7 concludes the thesis by summarizing the main contributions and implications of the work.

Chapter 2

Background and Related Work

2.1 Extended Reality for Procedural Training

Extended Reality (XR), encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), enables immersive and interactive learning environments that support procedural training, spatial instruction, and embodied interaction [16]. Compared to conventional instructional interfaces, XR systems can provide situated guidance directly within the user’s physical workspace, enabling contextualized learning and real-time interaction with both virtual and physical objects.

Recent research has demonstrated the effectiveness of XR for procedural training in domains including medical simulation, manufacturing, laboratory training, and industrial assembly [17, 18, 19]. In particular, MR environments allow instructional content to remain spatially aligned with real-world objects and user actions, supporting procedural understanding while preserving interaction with the physical environment.

Modern XR training systems increasingly move beyond static instruction delivery toward adaptive and personalized assistance capable of responding dynamically to user behavior and task context [20, 21]. This shift has been accelerated by recent advances in multimodal sensing, machine learning, and large language models integrated into immersive environments [22].

2.2 Multimodal Sensing in Mixed Reality

Mixed Reality environments provide access to rich multimodal interaction signals including gaze tracking, hand motion, head motion, speech, spatial mapping, and environmental context [23].

Compared to conventional desktop interaction, these sensing modalities provide substantially deeper behavioral information regarding user attention, interaction dynamics, spatial awareness, and procedural behavior.

Recent MR systems increasingly combine multiple sensing modalities to improve interaction understanding and adaptive feedback generation [24]. Gaze tracking has been widely used for attention estimation and interaction disambiguation, while hand tracking and gesture recognition support natural object manipulation and embodied interaction [25]. Spatial mapping and object tracking additionally enable alignment between virtual guidance and physical task environments.

Several recent systems have demonstrated the importance of multimodal sensor fusion for adaptive MR assistance. SIGMA [26] integrates gaze, gesture, voice, and positional tracking for mixed reality task assistance research, while XaiR [27] combines spatial sensing and LLM-based reasoning to support context-aware MR interaction. Recent surveys additionally emphasize that future adaptive MR systems will increasingly rely on multimodal AI models capable of jointly interpreting heterogeneous interaction streams in real time [28, 22].

2.3 Action Recognition, Interaction Understanding, and Task Tracking

Understanding user actions and procedural progress is a central requirement for adaptive procedural assistance systems. Existing approaches range from video-based action recognition and gesture classification to graph-based procedural reasoning and semantic state tracking.

Recent egocentric datasets such as Ego4D [29] and HoloAssist [30] have enabled significant progress in action understanding for immersive environments. In particular, HoloAssist introduced large-scale synchronized multimodal recordings of real-world procedural interaction together with annotations of user actions, instructor interventions, and procedural errors, highlighting the importance of fine-grained procedural reasoning and proactive assistance in XR environments.

Several recent systems have explored multimodal procedural assistance in MR environments. Autonomous Workflow for Multimodal Fine-Grained Training Assistants Towards Mixed Reality [31] combined multimodal sensing and vision-language reasoning for fine-grained task assistance, while SIGMA [26] provided an open-source platform for task assistance research integrating multimodal sensing and procedural overlays.

Although recent deep learning approaches achieve strong performance on large-scale action-

recognition benchmarks, many real-time XR procedural systems still rely on lightweight semantic state-tracking approaches due to deployment constraints, interpretability requirements, and limited domain-specific training data. Graph-based procedural reasoning additionally provides explicit dependency modeling and interpretable procedural state management that remain difficult to achieve using purely end-to-end recognition pipelines.

2.4 User-state Estimation in XR

Adaptive XR systems increasingly attempt to estimate user state in order to personalize guidance, adjust task difficulty, or trigger intervention strategies dynamically. Prior research has explored estimation of engagement, attention, workload, emotional state, and cognitive load using multimodal behavioral and physiological signals [32, 33].

Many existing cognitive-load estimation approaches rely heavily on physiological sensing modalities such as EEG, GSR, heart-rate monitoring, or facial analysis [23]. While these approaches can provide detailed measurements under controlled laboratory conditions, their deployment within practical mixed reality procedural environments remains challenging due to sensor intrusiveness, calibration complexity, and limited ecological validity.

Recent XR research has therefore increasingly shifted toward multimodal behavioral modeling using interaction dynamics available directly from immersive devices [34]. Explainable XR [35] demonstrated the use of LLM-assisted behavioral analysis for interpreting user interaction patterns in XR environments, while SMARTe-VR [33] explored adaptive VR learning using gaze, interaction logs, and biometric sensing.

Recent work by Stiber et al. [7] specifically examined confusion detection during physical procedural tasks using behavioral signals from gaze, hand movement, and head motion. Importantly, they operationalized confusion as a behaviorally observable state in which the user appears “stuck” and may require assistance, rather than as a direct measurement of an internal cognitive state. Their findings support the use of non-invasive behavioral interaction signals for estimating task-related difficulty during physical task execution.

Despite these advances, robust real-time user-state estimation in unconstrained MR procedural environments remains difficult. Generalization across users, tasks, and environments continues to represent a major open challenge, particularly for adaptive systems intended for deployment beyond controlled laboratory settings.

2.5 Large Language Models for Interactive XR Guidance

Recent advances in large language models (LLMs) have significantly influenced research on adaptive XR systems. LLMs enable dynamic natural-language interaction, contextual reasoning, procedural instruction generation, and adaptive conversational assistance within immersive environments [36, 22].

Several recent systems have explored integration of LLMs into XR interaction pipelines. Environment-aware conversational VR agents [37] demonstrated context-sensitive dialogue generation within immersive role-play environments, while XaiR [27] proposed a platform integrating LLM reasoning with real-world XR sensor data. Other work has explored conversational AI agents for industrial XR training and adaptive virtual tutoring [38].

At the same time, recent studies highlight several important limitations of current LLM-based XR systems, including hallucinated instructions, weak spatial grounding, insufficient real-time awareness, and limited robustness during long interactive sessions [39]. These limitations become particularly important in procedural task environments where incorrect guidance may negatively affect task execution or safety.

Consequently, recent research increasingly emphasizes context-aware prompting, multimodal grounding, and integration between procedural state tracking and language-model reasoning to improve reliability and adaptation quality in immersive assistance systems [22, 31].

2.6 Comparative Analysis of Existing Systems

Recent adaptive XR systems demonstrate several converging trends. First, multimodal sensing has become central to user interaction understanding and adaptive assistance generation [23]. Second, LLMs are increasingly used as reasoning engines for dynamic conversational guidance and contextual adaptation [22]. Third, XR systems are gradually moving from reactive query-based support toward proactive intervention strategies capable of anticipating user difficulties and procedural errors [30].

However, most existing systems still focus primarily on isolated aspects of adaptive XR interaction. Some emphasize conversational interaction without procedural state tracking [37], while others provide procedural overlays without robust user-state estimation or adaptive reasoning [26]. Many systems additionally rely on static workflows, limited personalization strategies, or domain-specific handcrafted logic that generalizes poorly across procedural environments.

Furthermore, few existing systems tightly integrate semantic procedural reasoning, multimodal behavioral modeling, and adaptive LLM guidance within a unified mixed reality runtime framework capable of operating in real-time procedural environments.

However, recent XR/AR procedural assistance systems address this problem from different directions. Systems such as Satori [8] emphasize proactive AR assistance and user intention modeling, while MRPilot [9] focuses on responsive procedural navigation in mixed reality. XR-Objects [10] emphasizes object-aware interaction and grounding, ProMemAssist [12] investigates working-memory-based intervention timing, and ProAgent [13] and Pro²Assist [11] explore proactive or step-aware assistance using multimodal egocentric reasoning. These systems demonstrate important progress toward adaptive and proactive assistance, but typically emphasize one primary axis of adaptation, such as intention inference, procedural navigation, object grounding, intervention timing, or multimodal reasoning.

In contrast, the proposed framework focuses on integrating semantic procedural task tracking with behavioral procedural difficulty estimation and runtime adaptive guidance in a real-time MR environment. The system is therefore positioned not as a generic XR guidance interface, but as a difficulty-aware adaptive procedural assistance framework. Its main distinction is that guidance is conditioned not only on what step the user is performing, but also on whether the user appears to be experiencing elevated procedural difficulty during task execution.

2.7 Research Gap Analysis and Design Requirements

The reviewed literature highlights several important research gaps motivating the proposed framework.

First, although multimodal sensing and adaptive assistance are increasingly common in XR systems, relatively few works combine semantic procedural tracking, temporal behavioral modeling, and adaptive language-model guidance within a unified real-time mixed reality architecture.

Second, many existing adaptive XR systems rely either on static rule-based adaptation or purely conversational interaction without explicit procedural state reasoning. As a result, these systems often lack robust awareness of procedural dependencies, off-task actions, or evolving execution state.

Third, user-state estimation in XR remains heavily dependent on intrusive physiological sensing or simplified laboratory conditions. Practical deployment-oriented mixed reality systems require approaches capable of inferring actionable behavioral states directly from naturally available

interaction signals.

Finally, current LLM-based XR assistants frequently suffer from hallucinated instructions, weak procedural grounding, and insufficient runtime awareness. Reliable adaptive guidance therefore requires tighter integration between semantic procedural state tracking, multimodal behavioral interpretation, and context-aware prompt construction.

Taken together, these limitations suggest that the remaining gap is not simply the absence of XR guidance, LLM assistance, or proactive intervention in isolation. Rather, the gap lies in combining these capabilities into a closed-loop MR framework that tracks procedural state over real physical objects, estimates observable procedural difficulty from behavioral signals, and adapts guidance during runtime based on both procedural context and inferred user difficulty.

Based on these observations, the proposed framework was designed around several core requirements:

- real-time semantic procedural state tracking;
- multimodal behavioral sensing using non-invasive MR interaction signals;
- behaviorally grounded procedural difficulty estimation for runtime adaptation;
- interpretable procedural reasoning and outcome classification;
- adaptive and proactive guidance generation;
- integration of LLM reasoning with explicit runtime procedural context;
- deployment feasibility within real mixed reality procedural environments.

These requirements collectively motivated the development of the adaptive mixed reality guidance framework proposed in this thesis.

Chapter 3

Methodology

3.1 Problem Formulation and System Overview

The goal of this thesis is to develop an adaptive mixed reality (MR) guidance framework capable of monitoring procedural task execution, interpreting user interaction behavior, and dynamically adapting assistance during real-world procedural tasks involving physical object interaction.

Conventional MR procedural systems typically rely on static instructional flows that provide identical guidance regardless of procedural deviations, user behavior, or evolving task difficulty. In contrast, the proposed framework formulates procedural assistance as a closed-loop adaptive process combining semantic task understanding, multimodal behavioral interpretation, and context-aware guidance generation.

The proposed framework consists of four tightly connected methodological components:

1. procedural task graph construction from natural-language protocols;
2. semantic procedural step tracking and outcome classification;
3. multimodal temporal modeling of observable procedural difficulty;
4. adaptive guidance generation conditioned on runtime procedural context.

Figure 4.1 illustrates the high-level methodological pipeline of the proposed framework.

The system continuously interprets multimodal interaction signals captured during mixed reality task execution, including object interactions, gaze behavior, hand motion, and head motion. These signals are transformed into semantic procedural actions aligned with a dependency-aware

procedural task graph. Runtime interaction behavior is additionally analyzed temporally to estimate elevated procedural difficulty states. The resulting procedural context and behavioral state information are then used to generate adaptive guidance through a context-aware large language model (LLM).

Unlike many prior adaptive MR systems that attempt to directly estimate latent cognitive load or affective state, the proposed framework intentionally focuses on observable procedural difficulty grounded in measurable interaction behavior. This design choice was motivated by the limited ecological validity of physiological cognitive-load estimation approaches in unconstrained MR environments and by the practical goal of enabling adaptive assistance using only non-invasive sensing modalities available directly on commodity MR hardware.

3.2 Procedural Task Graph Construction from Natural Language

Procedural tasks were represented as directed acyclic graphs (DAGs) generated from natural-language procedural descriptions as illustrated in Figure 4.3. The graph representation was selected because it enables interpretable dependency modeling, procedural state tracking, and explicit reasoning about procedural constraints during runtime execution.

Each node within the graph represented a semantic procedural step enriched with metadata including required objects, dependency constraints, optional substeps, and procedural outcomes. Directed edges encoded procedural relationships between steps.

Three dependency categories were defined throughout the framework:

- **Continuation dependencies**, representing sequential progression within procedural branches;
- **Component dependencies**, representing prerequisite preparation relationships;
- **Integration dependencies**, representing branch-merging operations where multiple intermediate components become combined into a subsequent procedural step.

The procedural graph served as the primary semantic representation used by the framework for procedural reasoning, valid-transition verification, and adaptive guidance generation.

Unlike linear instructional representations, the graph formulation additionally enabled reasoning about alternative execution paths and off-protocol actions. This allowed the framework to distinguish between expected actions, premature actions, repeated actions, recoverable deviations, and critical procedural violations according to the current dependency satisfaction state.

3.3 Semantic Step Tracking and Outcome Classification

The proposed framework continuously interpreted multimodal mixed reality interaction signals as semantic procedural actions relevant to the current procedural execution state.

Rather than relying on end-to-end video action recognition, the framework employed semantic procedural interpretation based on multimodal interaction constraints and dependency-aware procedural reasoning. Low-level interaction observations, including object interactions, gaze behavior, hand motion, and spatial relationships, were transformed into semantic procedural actions such as *grab*, *place*, *pour*, *stir*, and *press*.

Recognized actions were continuously aligned with the current procedural graph state to determine procedural progress and evaluate execution correctness relative to dependency constraints.

Outcome classification was conceptually inspired by the AMMA framework [3]. Each recognized action was classified into one of several semantic procedural outcome categories:

- **Expected**, representing correct actions satisfying current procedural dependencies;
- **Premature**, representing actions executed before prerequisite conditions were satisfied;
- **Repeat**, representing redundant repetition of previously completed actions;
- **Incorrect Recoverable**, representing non-critical procedural deviations;
- **Incorrect Unrecoverable**, representing critical procedural violations requiring corrective intervention.

This semantic procedural representation enabled interpretable procedural monitoring while supporting adaptive runtime guidance generation conditioned on procedural context and execution state.

3.4 Modeling Observable Procedural Difficulty from Multimodal Temporal Signals

The proposed framework estimated observable procedural difficulty from multimodal temporal interaction behavior captured during mixed reality task execution.

Unlike earlier stages of the project that framed the problem as cognitive-load and affective-state estimation, the final framework intentionally reformulated the problem as procedural difficulty inference grounded in observable behavioral interaction dynamics. This reformulation was

motivated by deployment realism, dataset limitations, and concerns regarding the validity of latent cognitive-state estimation in unconstrained MR environments. Moreover, this framing is consistent with recent work on confusion detection during physical tasks, where confusion is operationalized through observable behavioral evidence of being stuck or needing assistance rather than through direct measurement of subjective cognitive state.

Training data was derived from the HoloAssist dataset [30], which provides synchronized multimodal recordings together with procedural interaction annotations collected during real-world task execution. Labels representing stable versus elevated procedural difficulty states were constructed from procedural deviations and instructor-correction events rather than explicit cognitive-load measurements.

To model short-term interaction dynamics surrounding procedural deviations, the framework employed event-centered temporal modeling around annotated procedural events. Multimodal temporal interaction sequences were processed using a bidirectional gated recurrent unit (BiGRU) network [4] capable of modeling short-term temporal behavioral dynamics associated with unstable procedural interaction periods.

Task-identity features were intentionally excluded from the final training representation to reduce shortcut learning and improve generalization across unseen procedural tasks.

The resulting temporal signal was not interpreted as an absolute measurement of cognitive load or affective state. Instead, the inference output was treated operationally as an estimate of elevated procedural difficulty relative to stable procedural interaction behavior.

3.5 Adaptive Guidance Generation

The adaptive guidance module generated context-aware procedural assistance conditioned on runtime procedural context, semantic procedural outcomes, recent interaction history, and estimated procedural difficulty state.

Adaptive guidance generation was implemented using a locally hosted instruction-tuned large language model. Runtime prompts incorporated localized procedural context including the current procedural step, recently recognized actions and outcomes, available next procedural transitions, recent conversational history, and the estimated procedural difficulty state.

Multiple progressively richer prompt configurations were explored during development. Early prompt formulations relying primarily on static protocol descriptions frequently produced hallucinated or contextually irrelevant instructions because the model lacked awareness of the user’s

runtime execution state. Subsequent prompt designs instead focused on localized runtime procedural context, substantially improving contextual relevance and reducing hallucinated procedural guidance.

The final adaptive formulation additionally incorporated recent conversational history and procedural difficulty information. During elevated procedural difficulty periods, the assistant generated shorter, more directive, and action-focused guidance intended to minimize additional cognitive overhead during unstable interaction periods.

The framework additionally supported proactive intervention behavior. Rather than operating solely as a reactive question-answering assistant, the adaptive module could autonomously trigger guidance when repeated procedural errors accumulated, when dangerous procedural states occurred, or when elevated procedural difficulty persisted over time.

Generated guidance responses included both verbal procedural instructions and optional spatial object-highlighting cues integrated directly into the mixed reality environment.

3.6 Summary of the Proposed Method

The proposed methodology combines graph-based procedural reasoning, semantic procedural interpretation, multimodal temporal behavioral modeling, and adaptive LLM-based guidance generation into a unified mixed reality assistance framework.

The framework contributes three primary methodological ideas. First, procedural task execution is represented through interpretable DAG-based semantic state tracking generated from natural-language procedural descriptions. Second, observable procedural difficulty is inferred directly from multimodal interaction behavior without requiring intrusive physiological sensing or explicit cognitive-state estimation. Third, adaptive guidance generation combines procedural context, interaction history, and behavioral state information to support context-aware and proactive assistance during mixed reality task execution.

Together, these components establish a closed-loop adaptive MR guidance framework capable of monitoring procedural execution, identifying unstable interaction behavior, and dynamically adapting assistance according to evolving runtime procedural context.

Chapter 4

System Implementation

4.1 Runtime System Overview

The proposed adaptive mixed reality guidance framework was implemented as a distributed real-time mixed reality system deployed on Microsoft HoloLens 2. The runtime architecture combined a Unity-based mixed reality frontend deployed on the HoloLens 2 with multiple external Python-based backend services responsible for persistent object tracking, procedural reasoning, temporal user-state inference, and adaptive guidance generation. Unity served as the primary mixed reality application environment, while the Mixed Reality Toolkit 3 (MRTK3) provided hand tracking, gaze interaction, spatial interaction, and mixed reality user interface functionality throughout the system runtime.

Figure 4.2 illustrates the runtime architecture and communication flow between the deployed system components. The HoloLens 2 frontend continuously captured multimodal interaction signals including object detections, gaze input, hand interactions, head motion, and voice input. Spatial interaction events and procedural action streams were forwarded to the semantic procedural tracking pipeline, while multimodal behavioral features were transmitted to the temporal user-state inference module through lightweight UDP-based communication channels. When adaptive assistance conditions were satisfied, contextual task information, recent interaction history, and user-state information were sent to the LLM-based adaptive guidance server through a local HTTP interface. Generated guidance responses were then returned to the mixed reality frontend for visual and verbal feedback delivery.

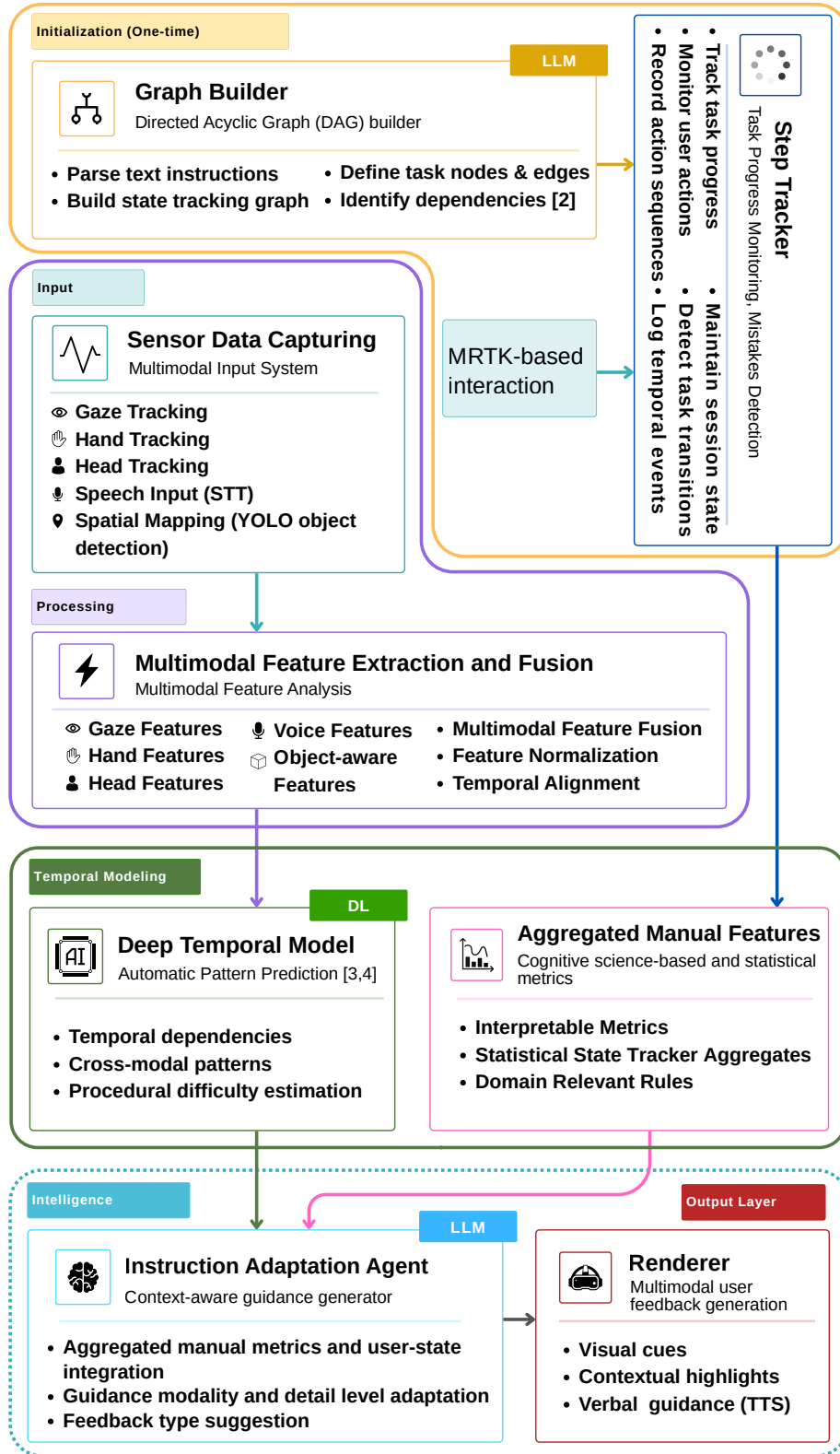


Figure 4.1: Runtime implementation architecture and data flow of the proposed adaptive mixed reality guidance framework.

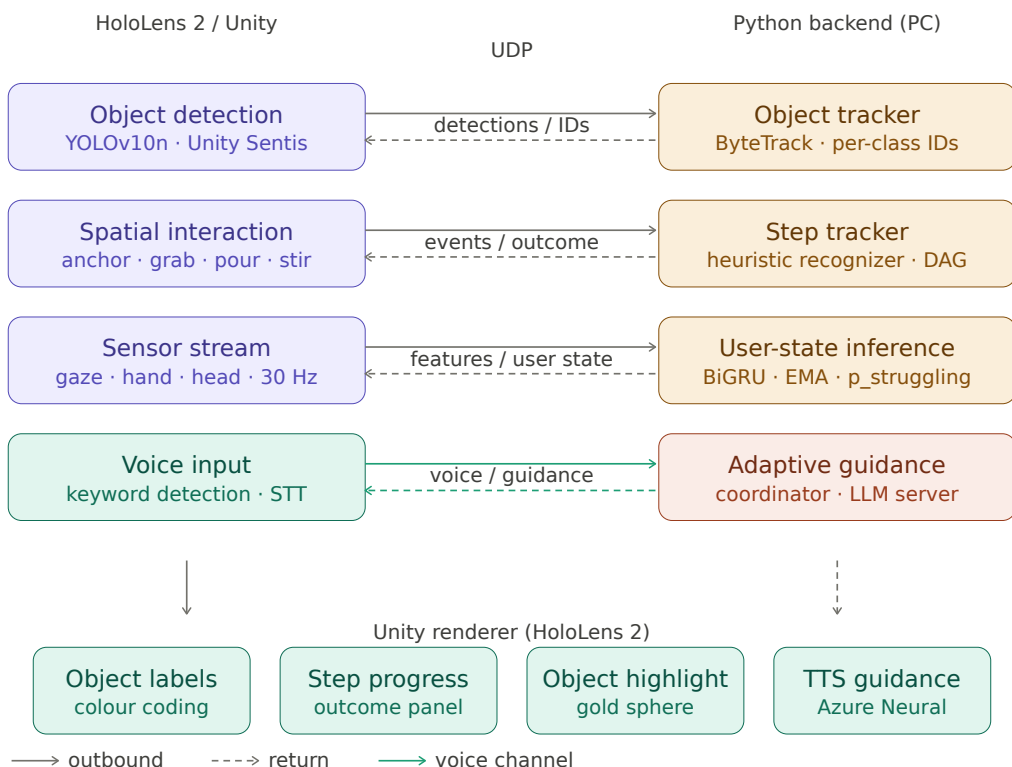


Figure 4.2: System pipeline for multimodal sensing, procedural interpretation, user-state inference, and adaptive guidance delivery.

4.2 Spatial Interaction and Object Tracking

4.2.1 YOLO-based Object Detection

Object detection was implemented using *HoloLens-YOLO-Object-Detection* project for HoloLens 2 developed by Dangberg et al. [1], which was adapted and extended to meet the procedural interaction requirements of the proposed system.

The original implementation used YOLOv8 models pretrained on the COCO dataset. However, several laboratory objects required for the chemistry-task condition were not sufficiently represented within the original training distribution. To improve robustness, additional object-detection models were fine-tuned using images collected directly from the experimental workspace using the HoloLens 2 device. The task-specific YOLO models achieved high object detection performance on the validation dataset ($mAP50 = 0.99$).

Separate detection models were maintained for the simple-transfer and chemistry-task conditions to improve runtime robustness under different procedural configurations. Fine-tuned models

were exported to ONNX format and deployed on-device through Unity Sentis for real-time inference.

Detected two-dimensional bounding boxes were projected into the three-dimensional workspace using sphere casts from the HoloLens camera origin through the center of each detection. The resulting world-space estimates were used to position object indicators, labels, and interaction colliders within the mixed reality scene.

4.2.2 Object Anchoring and Tracking Stabilization

Raw frame-by-frame object detections exhibited positional jitter and temporary identity instability during runtime interaction. To improve spatial consistency, detections were processed through a persistent tracking and anchoring pipeline.

To improve temporal consistency across detection frames, raw YOLO detections were forwarded through UDP to an external Python service running ByteTrack [2] in per-class tracking mode. ByteTrack assigned persistent track identities across frames using Kalman filtering and IoU-based association, allowing Unity to maintain stable `(track_id, class)` object identities throughout runtime interaction. This reduced object-switching artifacts that occurred when visually similar objects appeared near each other or when temporary detector instability caused intermittent class confusion. When ByteTrack was unavailable or no valid assignment was returned, the system fell back to class-based nearest-neighbor matching to preserve approximate object continuity. To prevent stale track persistence across sessions, the ByteTrack server automatically reset all active track identifiers after prolonged inactivity.

In addition to persistent tracking, the system employed a multi-stage object anchoring mechanism to stabilize world-space object positions. Initially, newly detected objects remained in an unanchored state in which three-dimensional positions were continuously estimated but not yet exposed to the interaction system. After repeated spatially consistent detections within a predefined tolerance threshold, the object transitioned into an anchored state and its world-space transform was frozen. Subsequent YOLO detections no longer directly updated the collider position during interaction. This prevented continuous positional drift caused by noisy frame-by-frame localization updates.

A lightweight visual indicator system was used to communicate runtime tracking states to the user. Anchored and interactable objects were represented using white sphere indicators positioned above the detected objects. When the user fixated on an object through gaze interaction, the

indicator changed to pink to communicate gaze selection and assist interaction disambiguation. During active manipulation, the indicator changed to green while the object was associated with the HandProxy interaction state.

Following a grab-and-release cycle, the object transitioned into a temporary lost state and was re-localized using fresh detections before re-entering the anchored state. This design introduced a short re-localization delay after placement events but eliminated most positional drift artifacts during manipulation and substantially improved interaction stability throughout procedural task execution.

4.2.3 Hand-object Interaction Processing

Hand tracking was implemented using MRTK3 skeletal hand joint data operating at approximately 30 Hz on the HoloLens 2. Grab and release states were estimated from combinations of hand posture, finger curl, wrist proximity to anchored objects, and short gaze-based disambiguation periods used to resolve nearby object ambiguity during interaction.

During object manipulation, direct YOLO-based tracking became unreliable because the user's hand frequently occluded the manipulated object from the HoloLens camera. To maintain stable interaction states during these periods, the system employed a HandProxy mechanism in which the manipulated object's transform temporarily followed the user's wrist joint rather than visual detections. This allowed the interaction system to preserve approximate object pose and interaction continuity throughout the manipulation phase despite temporary detector loss.

Additional contextual interaction features were continuously assembled during runtime, including wrist orientation, object tilt angle, pinch state, hand proximity, gaze fixation state, and relative spatial relationships between currently visible objects. Several semantic actions additionally relied on empirically tuned geometric constraints. For example, pouring interactions required sustained object tilt beyond approximately 45° toward a compatible target container, while grab interactions required the wrist joint to remain within an interaction radius of approximately 10–15 cm from an anchored object.

These contextual interaction signals were continuously transmitted to the backend semantic action recognition module for procedural interpretation.

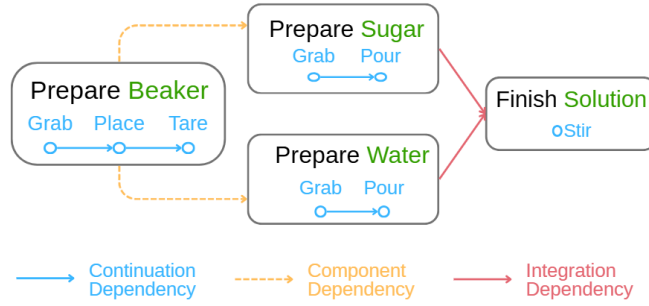


Figure 4.3: Example dependency-aware procedural task graph used for semantic step tracking. Blue arrows denote within-step continuation dependencies, dashed orange arrows represent component dependencies between procedural branches, and red arrows indicate integration dependencies where multiple procedural branches merge into a shared step.

4.3 Semantic Task Tracking

4.3.1 Runtime Procedural State Tracking

Procedural protocols generated through the methodology described in Chapter 3 were executed at runtime using a dependency-aware semantic state tracker.

Each runtime protocol instance maintained procedural step metadata, dependency relationships, valid transitions, and contextual error rules required for runtime procedural reasoning. Runtime interaction events were continuously matched against valid procedural transitions according to the current graph state.

The semantic tracker maintained active procedural branches, completed steps, dependency satisfaction states, and invalid-transition histories throughout runtime execution.

4.3.2 Semantic Action Recognition

Low-level interaction observations were continuously transformed into semantic procedural actions including *grab*, *place*, *pour*, *stir*, and *press*.

Action recognition relied on combinations of temporal interaction constraints, object states, hand-object proximity, object orientation, gaze fixation behavior, and spatial relationships between tracked objects. For example, pouring interactions required sustained object tilt beyond a predefined angular threshold toward a compatible target container, while placement interactions relied on proximity relationships between manipulated objects and predefined spatial regions.

Recognized semantic actions were forwarded to the procedural state tracker for dependency-aware runtime validation.

4.3.3 Outcome Classification and Runtime Broadcasting

Following runtime validation, each recognized semantic action was assigned to a procedural outcome category used by the adaptive guidance pipeline.

Procedural outcome events included expected actions, premature actions, repeated actions, recoverable deviations, and unrecoverable procedural violations. Outcome events were continuously broadcast to downstream runtime modules including the adaptive guidance pipeline and temporal user-state inference module.

4.4 User-state Inference Runtime

4.4.1 Feature Extraction

The user-state inference module estimated observable procedural difficulty from multimodal temporal interaction signals collected during mixed reality task execution. Importantly, the proposed system did not attempt to infer latent cognitive load, emotion, or affective state directly. Instead, the inference pipeline modeled comparatively stable procedural interaction periods versus interaction behavior associated with elevated procedural difficulty and observable task struggle.

The temporal inference model was trained using the publicly available HoloAssist dataset [6], which contains synchronized egocentric multimodal recordings including gaze, hand motion, head motion, RGB video, and instructional interaction annotations recorded during procedural task execution. The preprocessing pipeline utilized synchronized gaze, hand, and head-motion streams sampled at approximately 30 Hz together with timestamp-aligned procedural annotations collected across 25 task categories.

Binary labels were constructed using event-centered procedural annotations. Struggling windows were defined using events corresponding to verbally corrected wrong actions, uncorrected wrong actions, and instructor interventions associated with procedural correction. Stable windows were extracted from correct procedural actions while additionally enforcing a temporal exclusion region around nearby error-related events to reduce ambiguous transition states.

Rather than relying on fixed sliding windows, the preprocessing pipeline employed event-centered temporal segmentation. For each annotated event, a temporal window spanning approximately ± 2 seconds around the event timestamp was extracted, resulting in multimodal interaction sequences of approximately four seconds in duration. Windows containing fewer than 30 valid frames or excessive tracking dropout were discarded during preprocessing to reduce unreliable

sensor segments.

Two complementary feature representations were constructed. The first representation consisted of raw multimodal temporal sequences with shape (21787, 120, 30), containing 30 synchronized behavioral channels including gaze motion, hand dynamics, and head-motion signals. These raw temporal sequences were used directly by the recurrent inference model. The second representation consisted of aggregated statistical interaction features used for comparison against classical machine learning baselines. Aggregated features included hand velocity statistics, pause ratios, path efficiency, hover behavior, gaze motion entropy, and head stability metrics computed over each temporal window.

Task-family one-hot features were intentionally excluded from the final inference pipeline after preliminary analysis showed that they encouraged task-specific memorization behavior rather than generalizable behavioral modeling.

After preprocessing and filtering, the final dataset contained 21,787 temporal interaction windows collected from 1,720 procedural sessions spanning 25 task categories.

4.4.2 Runtime Calibration

Direct deployment of procedural difficulty probabilities across users proved unreliable because behavioral baselines varied substantially between individuals and experimental sessions. To improve runtime stability, the system employed a short per-session calibration phase before adaptive assistance was enabled.

During calibration, the inference module continuously estimated procedural difficulty probabilities while no adaptive interventions were generated. A participant-specific baseline distribution was then computed from the observed inference values. Runtime assistance triggering was subsequently performed relative to this calibrated baseline rather than absolute probability values alone.

To reduce rapid temporal fluctuations, the runtime inference signal was additionally smoothed using an exponential moving average ($\alpha = 0.3$). Adaptive intervention thresholds were initialized using approximately the 75th percentile of the calibration distribution.

4.4.3 Temporal Inference Runtime

Temporal user-state inference was implemented using a bidirectional gated recurrent unit (BiGRU) network [4] trained directly on raw multimodal temporal interaction sequences. The deployed architecture consisted of two bidirectional GRU layers with 64 and 32 hidden units respectively,

Table 4.1: Comparison of procedural difficulty inference models.

Model	Evaluation	Macro F1	Stable F1	Struggling F1
XGBoost	LOSO	0.673	0.782	0.564
BiGRU	Test split	0.847	0.908	0.785
BiGRU	LOTO	0.842	0.902	0.782

followed by a dense projection layer with dropout regularization ($p = 0.3$). The recurrent architecture was selected to model temporal behavioral dynamics directly from sequential multimodal interaction patterns rather than relying exclusively on manually engineered aggregate statistics.

To evaluate generalization across unseen procedural tasks, the primary evaluation protocol employed Leave-One-Task-Out (LOTO) cross-validation across all 25 HoloAssist task categories. As summarized in Table 4.1, the BiGRU model achieved a macro F1 score of 0.842 under LOTO evaluation, substantially outperforming an XGBoost [5] baseline trained on aggregated interaction features.

The similarity between standard train-test evaluation and LOTO evaluation performance suggested that the temporal model generalized beyond task-specific procedural patterns and instead captured transferable behavioral interaction dynamics associated with procedural difficulty.

During runtime deployment, the inference module continuously processed recent multimodal interaction sequences to estimate elevated procedural difficulty states in real time. The resulting procedural difficulty signal was continuously integrated into the adaptive guidance layer and used to trigger proactive assistance during periods associated with elevated procedural difficulty or unstable task execution behavior.

4.5 Adaptive Guidance Runtime

4.5.1 Voice Interaction

Voice interaction was implemented through a lightweight speech-triggered interaction pipeline integrated into the HoloLens 2 frontend. User speech was captured through the headset microphone and processed using keyword-triggered speech recognition.

Text-to-speech output was generated using Azure Cognitive Services Neural TTS [15]. Guidance responses were delivered both verbally and visually within the mixed reality environment.

4.5.2 LLM Prompt Construction and Inference

Adaptive guidance generation was implemented using a locally hosted instruction-tuned language model served through Ollama. The deployed model was `llama3.2:3b-instruct-q4_K_M`, a 4-bit quantized Llama 3.2 3B Instruct model. A small local model was selected to avoid cloud dependency during live task execution and to maintain low-latency inference within the experimental environment. The model was executed using a low temperature setting of 0.3 and a constrained generation budget to encourage concise, action-oriented responses.

Three prompt configurations were designed and evaluated to identify the production prompt template, following an incremental context-enrichment strategy inspired by Qorbani et al. [14]. Model A received the complete procedural protocol but no runtime execution state, representing a full-protocol baseline. Model B replaced the full protocol with selective runtime context, including the current procedural step, recent actions and outcomes, elapsed time on the current step, and available next procedural transitions. Model C additionally incorporated recent conversation history and the calibrated procedural difficulty state estimated by the BiGRU inference module.

Prompt quality was evaluated offline using an LLM-as-a-Judge methodology. A synthetic evaluation set consisting of 24 procedural scenarios was constructed across multiple protocol steps, procedural outcomes, and user-state conditions. Each prompt configuration generated responses for the same scenarios, and all response orders were permuted to reduce positional bias during judging. The judge model, `gemini-3-flash-preview`, evaluated responses according to correctness, contextual relevance, clarity, safety, actionability, and adaptation appropriateness.

Model C achieved the highest overall evaluation score of 4.19, compared with 3.19 for Model B and 1.98 for Model A. The largest improvement was observed in adaptation appropriateness, increasing from 1.90 for Model A to 4.17 for Model C, indicating that selective runtime context combined with procedural difficulty information substantially improved adaptive guidance quality.

Three progressively richer prompt configurations were designed and evaluated during development. The evaluated prompt templates and LLM-as-a-Judge evaluation protocol are provided in Appendix A.

4.5.3 Guidance Delivery and Runtime Intervention

Generated guidance responses were returned to the mixed reality frontend as structured outputs containing textual guidance and optional object-highlighting instructions.

Adaptive guidance could be triggered either explicitly through voice interaction or proactively by the system itself. Proactive intervention was activated when repeated procedural errors accumulated, when dangerous procedural states occurred, or when the estimated procedural difficulty remained elevated for a sustained temporal interval.

Guidance delivery additionally reused the runtime object-indicator system to provide spatially grounded visual feedback during adaptive assistance events.

4.6 Runtime Considerations

The distributed runtime architecture enabled separation between lightweight on-device interaction processing and computationally heavier backend inference modules.

Object detection and interaction processing were executed directly on the HoloLens 2 device to preserve responsive spatial interaction behavior, while temporal inference and adaptive language-model reasoning were executed externally on a connected workstation through a local wireless network.

This design reduced computational load on the headset while preserving real-time procedural interaction and adaptive guidance capabilities during runtime execution.

Chapter 5

Evaluation and Results

5.1 Evaluation Goals

The evaluation aimed to investigate the effectiveness of the proposed adaptive mixed reality guidance framework during procedural task execution. In particular, the study examined the reliability of semantic procedural tracking, the usefulness of multimodal procedural difficulty estimation, the quality of adaptive guidance generation, and the overall impact of adaptive assistance compared with a non-adaptive baseline system.

To address these goals, the evaluation combined technical system metrics, behavioral performance analysis, adaptive intervention analysis, subjective questionnaires, and qualitative feedback collected during controlled mixed reality task execution sessions.

5.2 Experimental Design

A controlled user study was conducted to evaluate the effectiveness of the proposed adaptive mixed reality guidance framework. The study compared a baseline procedural guidance system with the proposed adaptive guidance system across tasks with different levels of procedural complexity.

The experiment followed a counterbalanced mixed design involving two system conditions and two task types. Participants completed both a simple procedural task and a more complex chemistry-oriented procedural task. To reduce learning and order effects, the order of system conditions was counterbalanced across participants.

5.2.1 Participants

A total of 20 participants took part in the study. They were recruited from the university community and included graduate and undergraduate students with varying levels of prior experience with mixed reality systems and procedural laboratory tasks.

All participants received an introduction to the experimental procedure and HoloLens 2 interaction methods before the study began. Before the main experiment, they completed a short tutorial session to familiarize themselves with hand interactions, voice interaction, and the mixed reality interface.

5.2.2 Tasks and Experimental Conditions

Two procedural tasks with different levels of complexity were designed for the evaluation:

- **Simple Task:** A lower-complexity procedural task involving basic object manipulation and liquid transfer actions, with a small number of procedural dependencies and limited opportunities for procedural deviation. The task used simple, intuitive instructions with minimal domain-specific terminology and relatively low semantic interpretation demands.
- **Complex Task:** A higher-complexity procedural task involving sequential solution preparation and manipulation steps, increased procedural dependencies, domain-specific laboratory terminology, and a larger number of potential procedural deviations and recovery situations. Compared with the simple task, the chemistry-oriented task imposed higher procedural and semantic demands, requiring participants to interpret and execute more complex procedural instructions.

Two system conditions were evaluated:

- **Baseline Condition:** A matched non-adaptive version of the proposed framework. The baseline condition used the same procedural task model, task structure, semantic tracking pipeline, instructional content, and mixed reality interface as the adaptive condition. However, guidance was delivered as static procedural support based on the current task model and did not use runtime guidance adaptation, visual cue activation or voice-based guidance delivery.
- **Adaptive Condition:** The proposed adaptive mixed reality guidance system used the same procedural task model, semantic tracking pipeline, instructional content, and mixed reality

interface as the baseline condition, but additionally incorporated runtime procedural context and estimated procedural difficulty. Based on these signals, the system could adjust guidance timing, simplify terminology, provide context-specific support including voice-based guidance delivery, and activate visual object cues during task execution.

The baseline condition was designed as a matched non-adaptive version of the proposed framework rather than as an external state-of-the-art system. This choice was made to provide a controlled comparison between static guidance and runtime-adaptive guidance while keeping the procedural task model, semantic tracking pipeline, instructional content, and MR interface consistent across conditions.

Each participant completed two task sessions and experienced both system conditions and both task types. The four system–task combinations were distributed across two counterbalanced groups. Group A completed the simple task with the baseline system and the complex task with the adaptive system, whereas Group B completed the simple task with the adaptive system and the complex task with the baseline system. Thus, each participant experienced one baseline and one adaptive condition, as well as one simple and one complex task, while avoiding direct repetition of the same task under both systems. Because each participant did not complete all four system–task combinations, the design should be interpreted as **a counterbalanced mixed design** rather than a fully within-subject factorial design.

5.2.3 Study Procedure

Each experimental session consisted of four stages: introduction and tutorial, baseline calibration, procedural task execution, and post-task evaluation.

Before the main experiment, each participant completed the standard HoloLens 2 eye-gaze calibration procedure to ensure reliable gaze tracking and multimodal interaction sensing during task execution.

First, participants received an explanation of the experimental procedure and completed a short tutorial introducing HoloLens 2 interactions, object manipulation, voice interaction, and the mixed reality guidance interface.

Next, a short calibration phase was conducted to establish participant-specific baseline values for the user-state estimation module. During this phase, the system collected multimodal behavioral signals used to determine individualized procedural difficulty thresholds at runtime.

Participants then completed the assigned procedural tasks under the corresponding system conditions. During task execution, the system continuously performed semantic procedural tracking, monitored user interactions, and logged procedural events, behavioral signals, adaptive interventions, and runtime system information.

After each condition, participants completed post-task questionnaires, including the NASA-TLX workload assessment and guidance quality evaluation measures. At the end of the experiment, they also provided qualitative feedback on the usability, clarity, and perceived usefulness of the adaptive guidance system.

5.3 Evaluation Metrics

To evaluate the proposed adaptive mixed reality guidance framework comprehensively, the study combined technical system evaluation, behavioral performance analysis, adaptive guidance analysis, and subjective user evaluation. The selected metrics were designed to assess not only the technical reliability of the system, but also its influence on procedural task execution, user behavior, recovery from procedural deviations, and perceived usability during mixed reality interaction.

5.3.1 Technical System Performance Metrics

The technical system evaluation focused on the reliability and runtime responsiveness of the procedural tracking and adaptive guidance pipeline. These metrics were used to validate whether the system could accurately interpret procedural actions and provide real-time assistance during task execution.

The evaluated technical metrics included:

- **Semantic action recognition F1 score:** Evaluated the accuracy of semantic action detection during procedural task execution.
- **Outcome classification F1 score and confusion matrix:** Evaluated the reliability of procedural outcome classification, including expected, premature, repeated, and incorrect actions.
- **Step completion accuracy:** Measured the accuracy of procedural state tracking and the successful detection of completed procedural steps within the task graph.
- **Action-to-classification latency:** Measured the runtime delay between user interaction events and procedural outcome classification.

- **Error-to-feedback latency:** Measured the delay between procedural deviations and the delivery of corrective system feedback.
- **Voice query-to-response latency:** Measured the response time of the voice-based adaptive guidance pipeline.

5.3.2 Behavioral Performance Metrics

Behavioral performance metrics were used to evaluate how adaptive guidance influenced procedural task execution and observable user behavior during mixed reality interaction.

The primary behavioral metrics included:

- **Procedural error rate:** Measured the proportion of meaningful procedural actions that deviated from the expected workflow. The metric was computed as:

$$\text{Error Rate} = \frac{N_{\text{error}}}{N_{\text{expected}} + N_{\text{error}}} \quad (5.1)$$

where N_{error} includes premature, repeated, recoverable incorrect, and unrecoverable incorrect procedural actions. Harmless interactions were excluded from the analysis.

- **Struggling rate:** Measured the proportion of runtime windows classified as elevated procedural difficulty during task execution. The metric was computed as:

$$\text{Struggling Rate} = \frac{N_{\text{struggling windows}}}{N_{\text{total runtime windows}}} \quad (5.2)$$

where struggling windows were defined relative to participant-specific calibrated baseline thresholds obtained before the experimental session.

- **Recovery latency:** Measured the time required for participants to return to the expected procedural workflow after a procedural deviation. Recovery latency was computed as:

$$\text{Recovery Latency} = t_{\text{next expected}} - t_{\text{deviation}} \quad (5.3)$$

where $t_{\text{deviation}}$ represents the timestamp of a procedural deviation event and $t_{\text{next expected}}$ represents the timestamp of the next correctly executed expected procedural action. Consec-

utive deviation actions occurring before recovery were treated as part of the same deviation episode to avoid artificially inflating recovery counts.

For the validation analysis, normalized procedural difficulty values were compared between stable execution windows and deviation-centered windows:

$$\Delta D = D_{\text{deviation}} - D_{\text{stable}} \quad (5.4)$$

where $D_{\text{deviation}}$ represents the normalized procedural difficulty score computed from temporal windows immediately preceding procedural deviation events, and D_{stable} represents the normalized procedural difficulty score computed from temporal windows preceding correctly executed expected actions.

Both values were normalized relative to participant-specific calibrated baseline values obtained prior to the experiment. Positive values of ΔD therefore indicate elevated procedural difficulty before deviation events relative to stable task execution behavior.

For both conditions, procedural difficulty values were extracted from pre-action temporal windows spanning approximately 3 seconds before the corresponding procedural event in order to capture behavioral patterns leading up to expected actions and procedural deviations.

5.3.3 Adaptive Guidance Analysis Metrics

Additional analyses were conducted to evaluate the effectiveness of adaptive interventions during procedural task execution.

The adaptive guidance analysis metrics included:

- **Intervention success rate:** Measured the proportion of adaptive guidance interventions that were immediately followed by a correctly executed expected procedural action. The metric was computed as:

$$\text{Intervention Success Rate} = \frac{N_{\text{successful interventions}}}{N_{\text{total interventions}}} \quad (5.5)$$

where successful interventions represent adaptive guidance events followed by an expected procedural action without an additional intermediate deviation.

- **Post-intervention recovery latency:** Measured the time required for participants to return

to the expected procedural workflow after adaptive guidance was delivered. The metric was computed as:

$$\text{Post-Intervention Recovery Latency} = t_{\text{next expected}} - t_{\text{intervention}} \quad (5.6)$$

where $t_{\text{intervention}}$ represents the timestamp of adaptive guidance delivery and $t_{\text{next expected}}$ represents the timestamp of the next correctly executed expected procedural action.

These analyses were used to evaluate whether adaptive interventions contributed to procedural recovery and supported successful continuation of the task workflow.

5.3.4 Subjective User Evaluation Metrics

Subjective evaluation metrics were collected to assess users' perceptions of workload, guidance quality, and the overall interaction experience during mixed reality task execution.

The subjective evaluation consisted of:

- **NASA-TLX workload assessment:** Evaluated perceived workload across multiple dimensions, including mental demand, effort, and frustration. The overall NASA-TLX workload score was computed as:

$$\text{NASA-TLX}_i = \frac{M_i + E_i + F_i}{3} \quad (5.7)$$

where M_i , E_i , and F_i represent participant ratings for mental demand, effort, and frustration, respectively.

- **Guidance quality evaluation:** Perceived guidance quality was assessed in terms of clarity, confidence support, helpfulness, and overall guidance clarity during procedural task execution. The Guidance Quality Index (GQI) was computed as:

$$\text{GQI}_i = \frac{C_i + \text{Conf}_i + H_i + \text{GC}_i}{4} \quad (5.8)$$

where C_i , Conf_i , H_i , and GC_i represent questionnaire ratings for task-step clarity, confidence during task execution, guidance helpfulness, and explanation understandability, respectively.

In addition to quantitative questionnaires, participants provided qualitative feedback on adaptive guidance usefulness, interaction preferences, and the overall usability of the system.

5.4 Quantitative Results

5.4.1 Technical System Performance

The technical evaluation examined the reliability and runtime responsiveness of the proposed semantic procedural tracking and adaptive guidance framework during mixed reality task execution.

Semantic Action Recognition and Outcome Classification

The proposed semantic tracking pipeline continuously interpreted user interactions relative to the procedural task graph and classified actions into semantic procedural outcome categories, including expected, premature, repeated, and incorrect actions.

Across the evaluated sessions, semantic action recognition achieved an F1 score of 0.97 ± 0.03 , while procedural outcome classification achieved an F1 score of 0.96 ± 0.05 . These results indicate that the proposed semantic tracking framework reliably interpreted procedural interactions and distinguished between correct task progression and procedural deviations at runtime.

As shown in Figure 5.1, the aggregated outcome classification confusion matrix further demonstrated clear separation between expected procedural actions and deviation-related outcomes, with most remaining classification ambiguities occurring between semantically similar non-critical categories.

Step Completion Accuracy

Step completion accuracy was evaluated to determine whether the system could reliably track procedural progression within the task graph and correctly detect successful completion of procedural steps.

The proposed semantic tracking framework achieved a step completion accuracy of 1.00, indicating fully consistent procedural state progression across the evaluated sessions. Although minor ambiguities occasionally occurred during per-action outcome classification, they did not propagate into incorrect procedural graph states or step completion failures. The system maintained correct task progression even in the presence of repeated actions, premature interactions, and temporary procedural deviations.

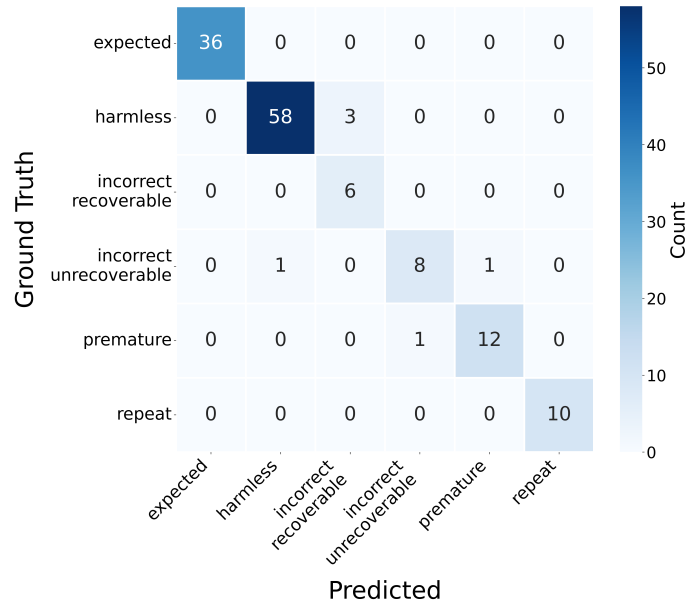


Figure 5.1: Confusion matrix for semantic procedural outcome classification.

These findings suggest that the graph-based procedural representation and semantic outcome reasoning approach enabled stable runtime task-state monitoring suitable for adaptive mixed reality guidance.

Runtime and Response Latency

Runtime latency analysis was conducted to evaluate whether the proposed system could provide sufficiently responsive real-time procedural guidance during task execution.

The mean action-to-classification latency was approximately 58 ± 12 ms, indicating near real-time semantic procedural monitoring during interaction. Error-to-corrective-feedback latency was measured at approximately 77 ± 65 ms, reflecting the additional reasoning required for deviation interpretation and corrective feedback generation.

Voice-based adaptive guidance exhibited higher end-to-end latency due to speech processing and LLM inference requirements. The mean latency between proactive or user-triggered voice queries and generated adaptive responses was approximately 4.8 ± 0.2 seconds. Although substantially higher than procedural tracking latency, the response time remained sufficiently responsive for procedural assistance during task execution.

Together, these results indicate that the proposed semantic tracking pipeline can reliably support real-time procedural reasoning and adaptive interaction within mixed reality environments, supporting RQ1.

Table 5.1: Summary of technical system performance across all evaluated sessions.

Metric	Mean	SD
Semantic action recognition F1	0.97	0.03
Outcome classification F1	0.96	0.05
Step completion accuracy	1.00	0.00
Action-to-classification latency	58 ms	12 ms
Error-to-corrective-feedback latency	77 ms	65 ms
Voice query-to-response latency	4.8 s	0.2 s

Note. F1 and accuracy values are reported as unitless proportions. Latency values are reported in milliseconds (ms) or seconds (s), depending on the pipeline stage.

5.4.2 Validation of Observable Procedural Difficulty Estimation

Elevated Procedural Difficulty Around Deviations

Before analyzing the behavioral effects of adaptive guidance, an additional validation analysis was conducted to examine whether elevated procedural difficulty estimates were associated with observable procedural deviations during task execution.

Normalized procedural difficulty values were compared between stable execution periods and windows preceding procedural deviation events. For each participant, procedural difficulty values were extracted from pre-action temporal windows spanning approximately 3 seconds before the corresponding procedural event.

The difference between both conditions was computed using Equation 5.4.

As shown in Fig. 5.2, normalized procedural difficulty scores were consistently higher before procedural deviations than during stable task execution periods. Across participants, the mean normalized procedural difficulty score increased from -0.002 ± 0.075 during stable execution windows to 0.063 ± 0.058 during deviation-centered windows, resulting in a mean difference of 0.065.

A Wilcoxon signed-rank test revealed a significant difference between both conditions ($W = 204.0, p < .001$). A paired t-test confirmed this result ($t(19) = 4.58, p < .001, d_z = 1.02$), indicating a large effect size.

These results suggest that elevated procedural difficulty estimates were consistently associated with procedural deviations during task execution, supporting the use of the procedural difficulty signal as an operational indicator for adaptive mixed reality guidance assistance, addressing RQ2.

5.4.3 Behavioral Impact of Adaptive Guidance

Procedural Error Rate

To evaluate whether adaptive guidance reduced procedural deviations during task execution, procedural error rates were analyzed across all experimental conditions using a linear mixed-effects model (LMM). The model included *System* (Baseline vs. Adaptive), *Task* (Simple vs. Complex), and their interaction as fixed effects, with participant included as a random intercept.

$$\text{ErrorRate}_{ij} = \beta_0 + \beta_1 \text{System}_{ij} + \beta_2 \text{Task}_{ij} + \beta_3 (\text{System} \times \text{Task})_{ij} + u_i + \epsilon_{ij} \quad (5.9)$$

where β_0 represents the intercept, β_1 and β_2 represent the fixed effects of system condition and task complexity, β_3 represents the interaction effect between both factors, u_i represents the participant-specific random intercept, and ϵ_{ij} represents residual error.

Figure 5.3 shows the distribution of procedural error rates across all four experimental conditions, while Fig. 5.4 illustrates the interaction effect between system condition and task complexity.

Under the simple task condition, procedural error rates remained comparatively low for both systems. The estimated marginal mean procedural error rate for the baseline simple condition was approximately 0.272, while the adaptive simple condition showed a comparable estimated marginal mean of approximately 0.247. The linear mixed-effects analysis revealed no significant effect of adaptive guidance during the simple task condition ($\beta = -0.0249, p = .619$).

In contrast, substantially higher procedural error rates were observed during the chemistry-oriented task. The estimated marginal mean procedural error rate increased to approximately 0.633 under the baseline chemistry condition, whereas the adaptive chemistry condition showed a substantially lower estimated marginal mean of approximately 0.322.

The linear mixed-effects analysis revealed a significant main effect of task complexity ($\beta = 0.3610, p < .001$), indicating that chemistry-oriented tasks produced substantially higher procedural error rates overall compared to the simple task condition.

More importantly, a significant *System* $\times$ *Task* interaction effect was observed ($\beta = -0.2871, p < .001$), indicating that the behavioral impact of adaptive guidance depended strongly on task complexity. As illustrated in Fig. 5.4, procedural error rates increased sharply from the simple to chemistry task under the baseline condition, whereas the adaptive condition showed a substantially smaller increase.

Table 5.2: Linear mixed-effects model results for procedural error rate analysis.

Term	β	p -value
Intercept (Baseline/Simple)	0.2719	< .001***
System [Adaptive vs. Baseline]	-0.0249	0.6187
Task [Complex vs. Simple]	0.3610	< .001***
System $\times$ Task	-0.2871	< .001***

These findings indicate that the proposed adaptive guidance framework improved procedural workflow stability during higher-complexity chemistry-oriented tasks by reducing procedural deviations and incorrect task progression.

Struggling Rate Analysis

To evaluate whether adaptive guidance reduced sustained procedural difficulty during task execution, struggling rates were analyzed across all experimental conditions using a linear mixed-effects model (LMM). The model included *System* (Baseline vs. Adaptive), *Task* (Simple vs. Complex), and their interaction as fixed effects, with participant included as a random intercept:

$$\text{StrugglingRate}_{ij} = \beta_0 + \beta_1 \text{System}_{ij} + \beta_2 \text{Task}_{ij} + \beta_3 (\text{System} \times \text{Task})_{ij} + u_i + \epsilon_{ij} \quad (5.10)$$

where β_0 represents the intercept, β_1 and β_2 represent the fixed effects of system condition and task complexity, β_3 represents the interaction between these factors, u_i represents the participant-specific random intercept, and ϵ_{ij} represents the residual error.

Figure 5.5 shows the distribution of struggling rates across all four experimental conditions, while Fig. 5.6 illustrates the corresponding interaction effect between system condition and task complexity.

Under the simple task condition, struggling rates remained comparatively low for both systems. The estimated marginal mean struggling rate for the baseline simple condition was 0.0646, while the adaptive simple condition showed a comparable value of 0.0735. The LMM analysis revealed no significant effect of adaptive guidance during the simple task condition ($\beta = 0.0089, p = .848$).

In contrast, substantially higher struggling rates were observed during the chemistry-oriented task. The estimated marginal mean struggling rate increased to 0.6988 under the baseline chemistry condition, whereas the adaptive chemistry condition showed a reduced estimated marginal

Table 5.3: Linear mixed-effects model results for struggling rate analysis.

Term	β	p -value
Intercept (Baseline/Simple)	0.0646	0.0489*
System [Adaptive vs. Baseline]	0.0089	0.8479
Task [Complex vs. Simple]	0.6342	< .001***
System $\times$ Task	-0.2324	< .001***

mean of 0.4753.

The linear mixed-effects analysis revealed a significant main effect of task complexity ($\beta = 0.6342, p < .001$), indicating that chemistry-oriented tasks produced substantially higher struggling rates overall compared to the simple task condition.

More importantly, a significant *System $\times$ Task* interaction effect was observed ($\beta = -0.2324, p < .001$), indicating that the behavioral impact of adaptive guidance depended strongly on task complexity. As illustrated in Fig. 5.6, struggling rates increased sharply from the simple to chemistry task under the baseline condition, whereas the adaptive condition showed a substantially smaller increase.

These findings suggest that adaptive guidance was particularly effective under conditions involving higher procedural and semantic complexity, where users were more likely to experience sustained procedural difficulty during task execution. These findings support RQ4.

Recovery Latency Analysis

Recovery latency was analyzed to assess how quickly participants returned to the expected procedural workflow after a procedural deviation during higher-complexity task execution.

Because the previous analyses showed minimal behavioral differences between system conditions during the low-complexity simple task, this analysis focused on the chemistry-oriented conditions, where substantial differences in struggling rates and procedural error rates were observed.

Recovery latency was computed according to Equation 5.3 as the elapsed time between the onset of a procedural deviation and the next correctly executed expected procedural action. Each participant contributed one session-level median recovery latency value.

Figure 5.7 shows the distribution of recovery latency values for the baseline and adaptive chemistry-oriented conditions. Participants in the adaptive condition recovered substantially faster after procedural deviations than participants in the baseline condition.

As summarized in Table 5.4, the adaptive chemistry-oriented condition produced a mean re-

Table 5.4: Recovery latency comparison between baseline and adaptive chemistry-oriented conditions.

Condition	n	Mean (s)	SD (s)	Median (s)	Range (s)
Adaptive Chem	8	56.46	24.45	60.93	21.14–86.10
Baseline Chem	10	132.22	50.23	120.08	51.77–217.59

covery latency of 56.46 ± 24.45 seconds (median = 60.93 s), whereas the baseline chemistry-oriented condition showed a substantially higher mean latency of 132.22 ± 50.23 seconds (median = 120.08 s).

A Mann–Whitney U test revealed a significant difference between the two conditions ($U = 5, p = .0009$). Cliff’s $\delta = -0.875$ indicated a large effect size, suggesting substantially faster procedural recovery under adaptive guidance.

These findings suggest that adaptive guidance not only reduced procedural deviations and sustained procedural difficulty during higher-complexity task execution, but also helped participants recover more efficiently after deviations occurred.

5.4.4 Post-hoc Analysis of Adaptive Guidance

Intervention Success Analysis

An additional post-hoc analysis was conducted to evaluate whether adaptive guidance interventions were followed by successful procedural recovery and continuation of the intended workflow.

Intervention success rate was computed according to Equation 5.5 as the proportion of adaptive interventions that were immediately followed by a correctly executed expected procedural action without an additional intermediate deviation.

Across the evaluated adaptive chemistry-oriented sessions, the mean intervention success rate was **74.60%** ($SD = 7.60\%$), with a median of **77.80%**. Observed success rates ranged from **62.50%** to **83.30%**, indicating that most adaptive interventions were followed by successful procedural recovery.

These findings suggest that adaptive guidance interventions frequently supported successful procedural continuation after procedural deviations or elevated procedural difficulty states.

Post-intervention Recovery Analysis

Post-intervention recovery analysis examined how quickly participants resumed correct procedural execution after adaptive guidance was delivered.

Post-intervention recovery latency was computed according to Equation 5.6 as the elapsed time between adaptive guidance delivery and the next correctly executed expected procedural action.

Across the evaluated adaptive chemistry-oriented sessions, the mean post-intervention recovery latency was approximately 11–14 seconds, while the median latency ranged from approximately 9–11 seconds.

These findings suggest that participants typically resumed correct procedural execution shortly after receiving adaptive guidance, supporting the effectiveness of the proposed adaptive intervention mechanism during higher-complexity task execution. However, post-intervention recovery latency inherently included the time required for participants to listen to or read the adaptive guidance before resuming the task. Therefore, the reported latency values reflect overall behavioral recovery time following adaptive intervention rather than isolated motor or decision-making response time alone.

Moreover, these findings suggest that adaptive guidance generated from procedural context, semantic outcomes, and procedural difficulty information can provide contextually appropriate assistance during real-world mixed reality task execution, addressing RQ3.

Representative Adaptive Guidance Examples

Several runtime interaction cases demonstrated the adaptive system’s ability to provide context-aware procedural assistance during complex mixed reality task execution. Table 5.5 presents representative examples illustrating semantic interpretation of ambiguous user queries, adaptive explanation simplification, and context-aware procedural safety guidance.

These examples demonstrate that the proposed adaptive framework was capable of integrating procedural context, multimodal interaction state, and natural-language input into coherent real-time guidance behavior. In addition to responding to explicit user requests, the system dynamically interpreted imperfect speech input, simplified scientific terminology, generated safety-oriented responses, and adapted procedural explanations according to the participant’s current task state.

5.5 Subjective User Evaluation

5.5.1 NASA-TLX Workload Analysis

Subjective workload was evaluated using the NASA-TLX Core score, computed according to Equation 5.7. To examine the effects of adaptive guidance and task complexity on perceived work-

Scenario	Observed Input / Context	Adaptive System Response
Semantic interpretation of mistranscribed scientific terminology	Current procedural step: <i>"Dispense diluent into the mixing beaker."</i> User query: <i>"What is dilutant?"</i> The spoken term was incorrectly transcribed by the speech recognizer.	Despite the mistranscription, the adaptive system correctly inferred that the participant was referring to the procedural term "diluent" and generated the response: <i>"Diluent, in this case, is just water. We use it to mix with the solution."</i> The generated response additionally highlighted the water bottle and beaker inside the MR environment, helping the participant associate unfamiliar terminology with physical objects.
Robust handling of imperfect speech transcription	The participant asked: <i>"Can I drink the liquid in the bigger?"</i> The speech recognizer incorrectly transcribed "beaker" as "bigger".	Instead of producing unrelated output, the adaptive system inferred the intended object reference and generated the safety-oriented response: <i>"No, you should not drink the liquid in the beaker. It is a colored solution used for the experiment and may contain chemicals."</i> The system additionally highlighted the beaker object in the MR scene.
Adaptive step decomposition during procedural confusion	The participant demonstrated prolonged struggling while attempting to perform the liquid transfer procedure and asked an unclear follow-up question during execution.	Instead of repeating the original protocol description, the system decomposed the task into smaller actionable instructions: <i>"First, pick up the water bottle. Hold the beaker steady with your other hand and slowly tilt the bottle."</i> The generated response adapted dynamically to the participant's current interaction state and reduced procedural complexity into sequential executable actions.

Table 5.5: Representative examples of adaptive mixed reality guidance behavior during runtime interaction.

load, NASA-TLX Core scores were analyzed across all experimental conditions using a linear mixed-effects model (LMM). The model included *System* (Baseline vs. Adaptive), *Task* (Simple vs. Complex), and their interaction as fixed effects, with participant included as a random intercept.

Figure 5.8 shows the distribution of NASA-TLX Core scores across the four experimental conditions, while Fig. 5.9 presents the corresponding workload subscale distributions.

Under the simple task condition, perceived workload remained comparatively low for both systems. The estimated marginal mean NASA-TLX Core score for the baseline simple condition was approximately 1.80, and the adaptive simple condition showed a comparable estimated marginal mean. The linear mixed-effects analysis revealed no significant effect of adaptive guidance during the low-complexity simple task condition ($\beta = 0.0000, p = 1.000$).

In contrast, substantially higher workload scores were observed during the complex task under the baseline condition. The estimated marginal mean NASA-TLX Core score increased by

Table 5.6: Linear mixed-effects model results for NASA-TLX Core analysis.

Term	β	p -value
Intercept (Baseline/Simple)	1.8000	< .001***
System [Adaptive vs. Baseline]	0.0000	1.0000
Task [Complex vs. Simple]	6.3333	< .001***
System $\times$ Task	-4.4667	< .001***

approximately 6.33 points relative to the simple task condition ($\beta = 6.3333, p < .001$). However, the adaptive complex condition showed a substantially smaller increase in perceived workload.

Most importantly, the analysis revealed a significant *System $\times$ Task* interaction effect ($\beta = -4.4667, p < .001$), indicating that the effect of adaptive guidance depended strongly on task complexity. As illustrated in Fig. 5.8, adaptive guidance substantially reduced perceived workload during higher-complexity task execution, while producing comparatively limited subjective differences during the simple task.

The subscale analysis further showed that the largest reductions occurred in perceived effort and frustration, while mental demand also decreased under the adaptive complex condition compared to the baseline complex condition.

These findings suggest that adaptive guidance not only improved objective procedural performance, but also substantially reduced perceived workload and procedural difficulty during higher-complexity task execution.

5.5.2 Perceived Guidance Quality and Usefulness

Perceived guidance quality was evaluated using the Guidance Quality Index (GQI) computed according to Equation 5.8. To examine the effects of adaptive guidance and task complexity on perceived guidance usefulness, GQI scores were analyzed across all experimental conditions using a linear mixed-effects model (LMM). The model included *System* (Baseline vs. Adaptive), *Task* (Simple vs. Chemistry), and their interaction as fixed effects, with participant included as a random intercept.

Figure 5.10 shows the distribution of Guidance Quality Index scores across all four experimental conditions and illustrates the interaction between system condition and task complexity.

Under the simple task condition, perceived guidance quality remained comparatively high for both systems. The estimated marginal mean GQI score for the baseline simple condition was approximately 6.55, while the adaptive simple condition showed a comparable estimated marginal mean of approximately 6.40. The linear mixed-effects analysis revealed no significant effect of

Table 5.7: Linear mixed-effects model results for Guidance Quality Index (GQI) analysis.

Term	β	p -value
Intercept (Baseline/Simple)	6.5500	< .001***
System [Adaptive vs. Baseline]	-0.1500	0.5803
Task [Chemistry vs. Simple]	-4.4250	< .001***
System $\times$ Task	4.2667	< .001***

adaptive guidance during the low-complexity simple task condition ($\beta = -0.1500, p = .5803$).

In contrast, substantially lower guidance quality scores were observed during the chemistry-oriented task under the baseline condition. Relative to the baseline simple condition, the baseline chemistry-oriented condition showed a large decrease in perceived guidance quality ($\beta = -4.4250, p < .001$). However, the adaptive chemistry-oriented condition demonstrated a substantial improvement in perceived guidance quality compared to the baseline chemistry-oriented condition.

More importantly, the analysis revealed a significant *System $\times$ Task* interaction effect ($\beta = 4.2667, p < .001$), indicating that the perceived usefulness and effectiveness of guidance depended strongly on task complexity. As illustrated in Fig. 5.10, guidance quality ratings remained consistently high under the adaptive chemistry-oriented condition, whereas substantially lower ratings were observed under the baseline chemistry-oriented condition.

These findings suggest that adaptive guidance substantially improved perceived guidance usefulness, clarity, and overall support effectiveness during higher-complexity chemistry-oriented task execution, while producing comparatively limited subjective differences during low-complexity procedural tasks.

5.6 Qualitative Feedback and Observations

Qualitative participant feedback generally aligned with the quantitative findings obtained during the evaluation. Participants reported that the adaptive guidance system was particularly helpful during the chemistry-oriented task, where unfamiliar terminology and increased procedural complexity produced higher perceived difficulty compared to the simple task condition.

Several participants noted that context-aware explanations and visual object highlighting helped them associate procedural instructions with unfamiliar laboratory objects more effectively. Visual hints were frequently described as the most useful guidance modality during procedural execution.

Participants additionally reported that the adaptive system reduced uncertainty and stress dur-

ing difficult task steps because guidance responses adapted dynamically to the current procedural context rather than statically repeating instructions. Some participants specifically mentioned that the presence of the assistant reduced anxiety during higher-complexity task execution.

Participants also expressed different preferences regarding voice and visual guidance modalities. While several participants preferred reading instructions directly from the mixed reality interface due to the slower pace of spoken responses, short voice-based notifications such as step completion confirmations and danger alerts were frequently described as highly effective during active task execution.

Overall, the qualitative observations supported the quantitative findings by suggesting that adaptive mixed reality guidance was particularly beneficial during higher-complexity procedural tasks involving unfamiliar terminology, multi-step coordination, and increased procedural uncertainty.

5.7 Summary of Findings

The evaluation results demonstrated that the proposed adaptive mixed reality guidance framework successfully integrated semantic procedural tracking, observable procedural difficulty estimation, and context-aware adaptive guidance into a coherent real-time assistance system.

Technical evaluation showed that the semantic tracking pipeline achieved high procedural interpretation performance, including strong semantic action recognition accuracy, reliable procedural outcome classification, and low-latency runtime responsiveness. The proposed procedural difficulty estimation approach additionally demonstrated significantly elevated difficulty estimates around procedural deviation events, supporting its use as an operational indicator of observable procedural struggle during task execution.

Behavioral analyses further showed that adaptive guidance produced comparatively limited effects during the low-complexity simple task condition, but substantially improved performance during higher-complexity chemistry-oriented task execution. In particular, the adaptive condition demonstrated reduced struggling rates, lower procedural error rates, and substantially faster recovery following procedural deviations.

Subjective evaluation results similarly indicated that adaptive guidance reduced perceived workload and improved perceived guidance quality during higher-complexity task execution. Significant interaction effects observed across multiple behavioral and subjective analyses suggested that the effectiveness of adaptive guidance depended strongly on procedural complexity and par-

ticipant difficulty during task execution.

Post-hoc analyses and qualitative observations further demonstrated that the adaptive framework was capable of generating context-aware procedural assistance, simplifying unfamiliar terminology, interpreting imperfect speech input, and dynamically adapting procedural explanations according to the participant's current interaction state.

Overall, the findings suggest that adaptive mixed reality guidance systems integrating semantic procedural understanding, multimodal behavioral analysis, and real-time natural-language assistance may provide substantial benefits for complex procedural task support in mixed reality environments.

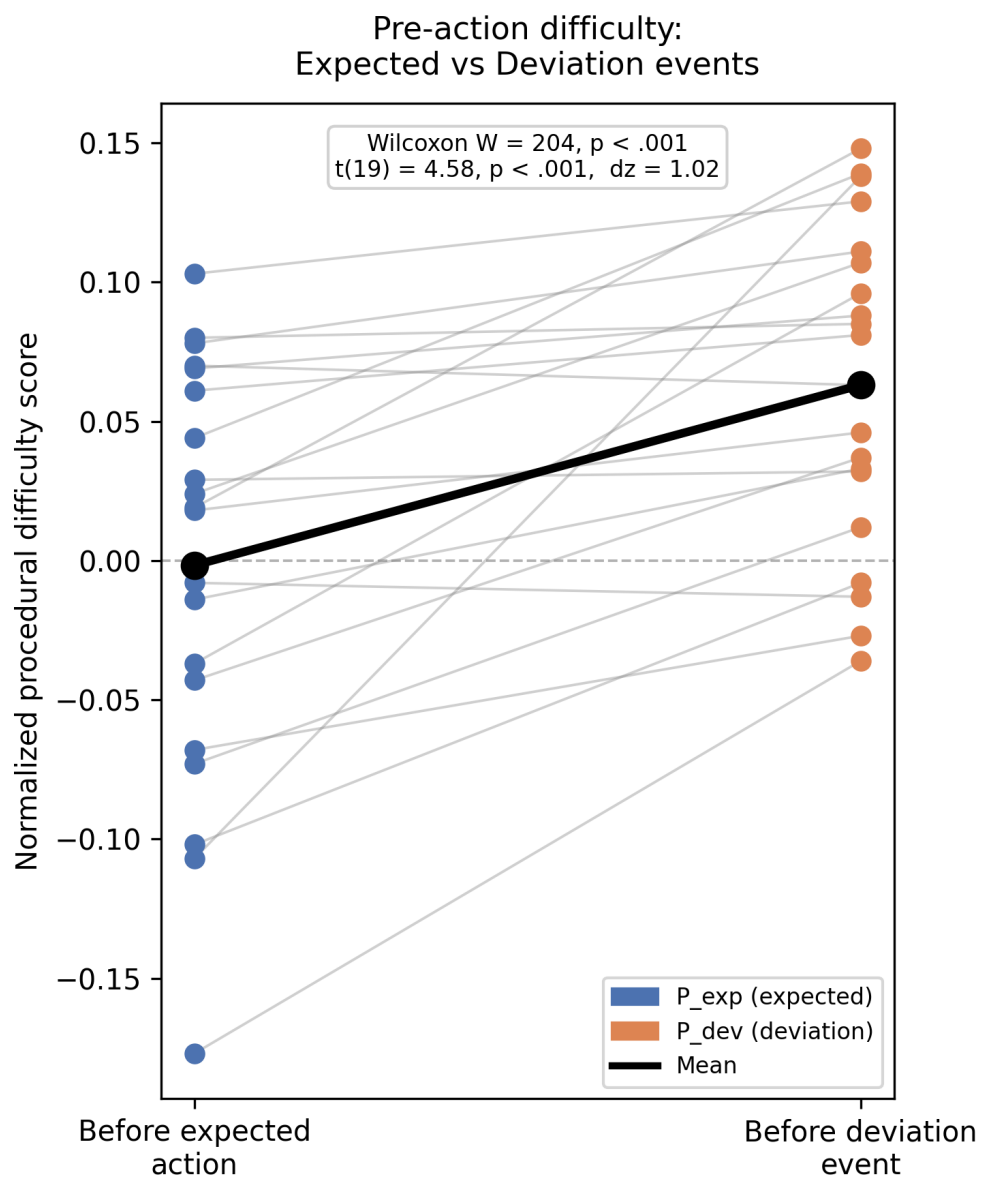


Figure 5.2: Normalized procedural difficulty before expected actions and procedural deviations.



Figure 5.3: Distribution of procedural error rates across baseline and adaptive guidance conditions for simple and chemistry-oriented tasks.

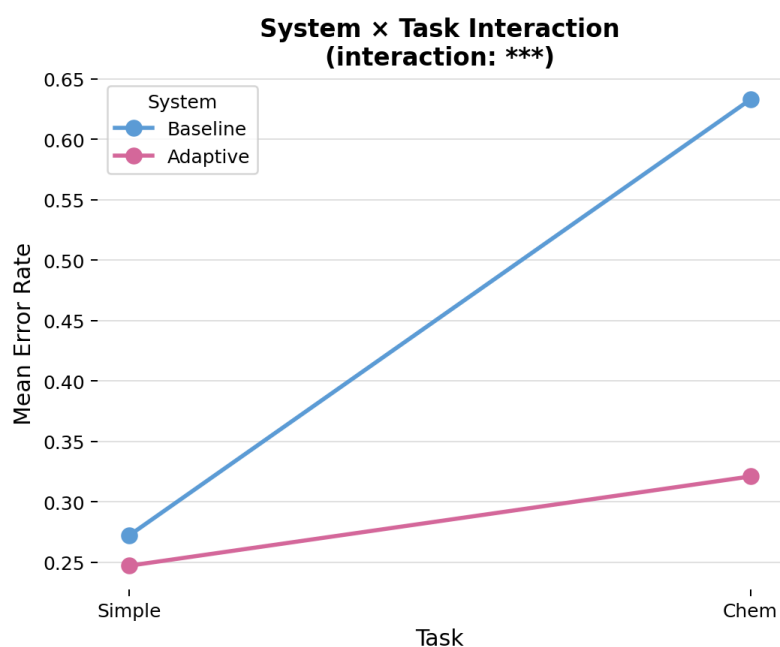


Figure 5.4: Interaction effect between system condition and task complexity for procedural error rate.

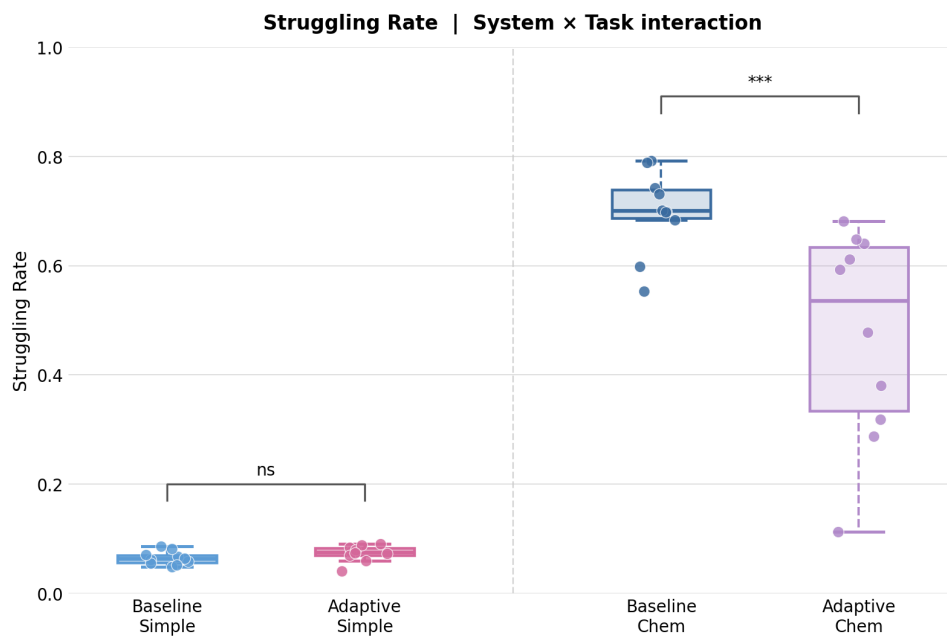


Figure 5.5: Distribution of struggling rates across baseline and adaptive guidance conditions for simple and chemistry-oriented tasks.

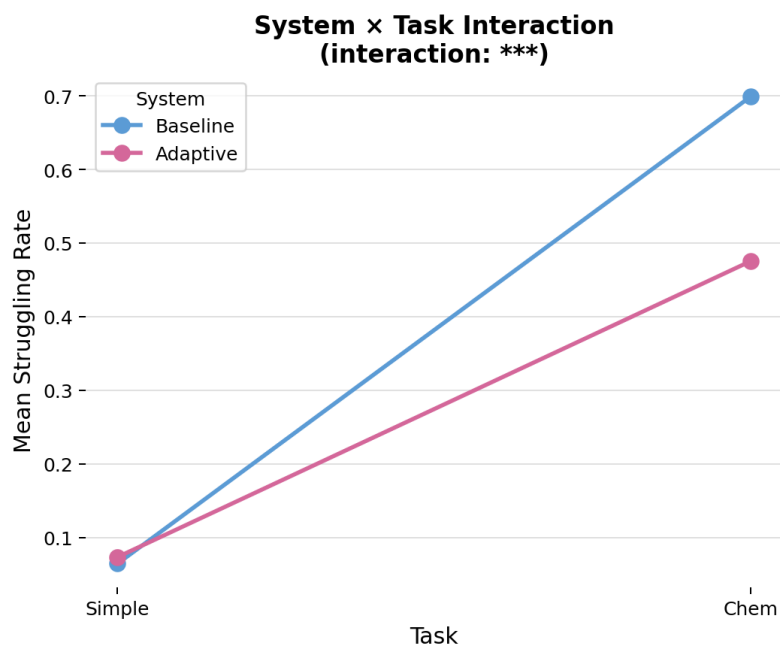


Figure 5.6: Interaction effect between system condition and task complexity for struggling rate.

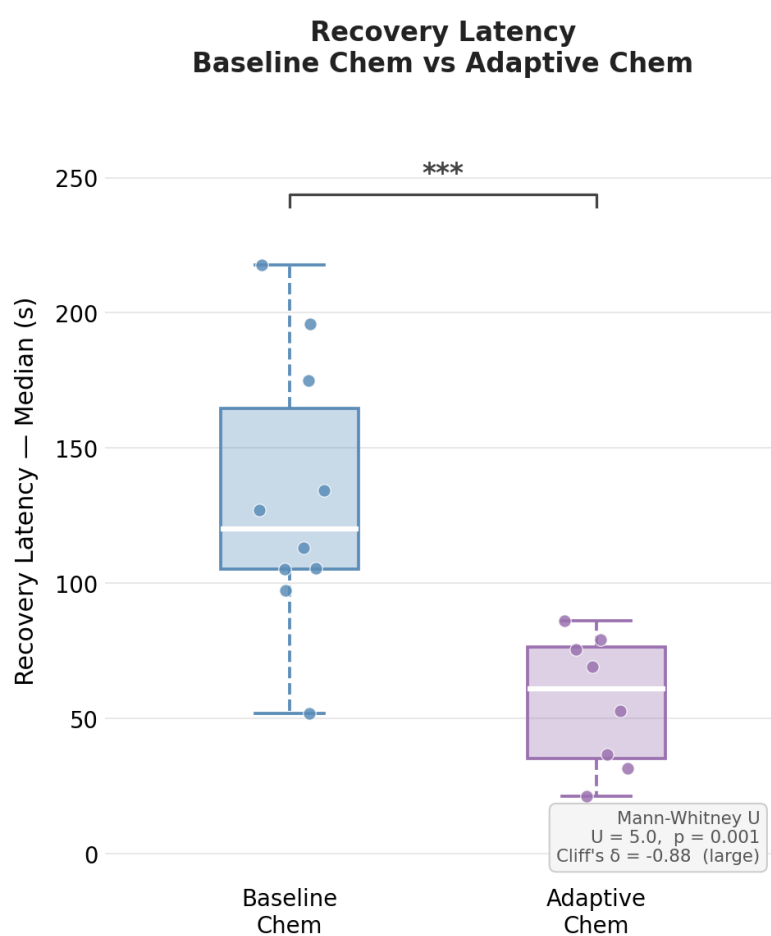


Figure 5.7: Recovery latency comparison between baseline and adaptive guidance conditions during chemistry-oriented task execution.

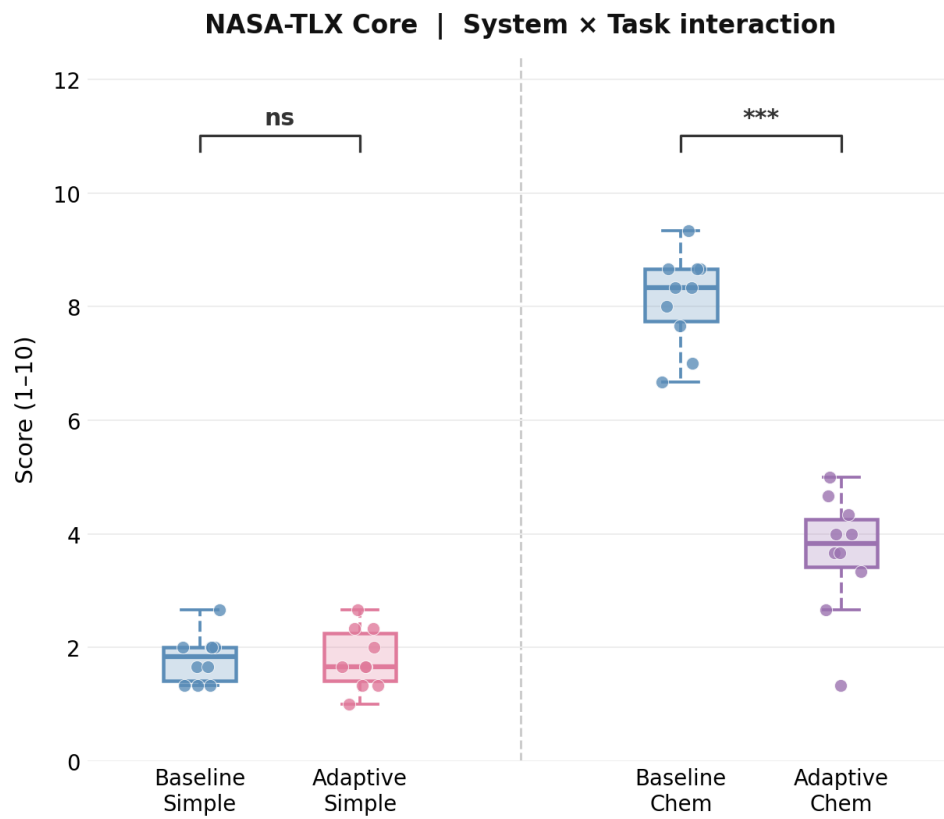


Figure 5.8: NASA-TLX Core scores across baseline and adaptive guidance conditions for simple and complex tasks.

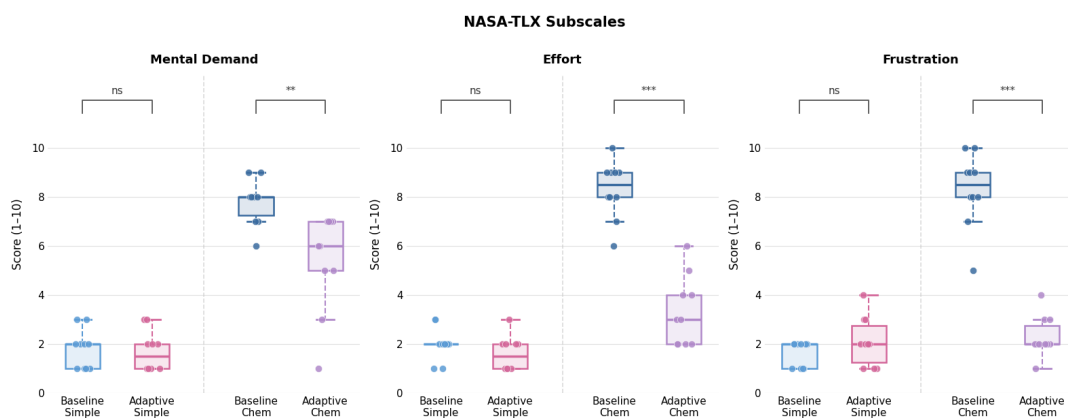


Figure 5.9: NASA-TLX workload subscale distributions across experimental conditions.

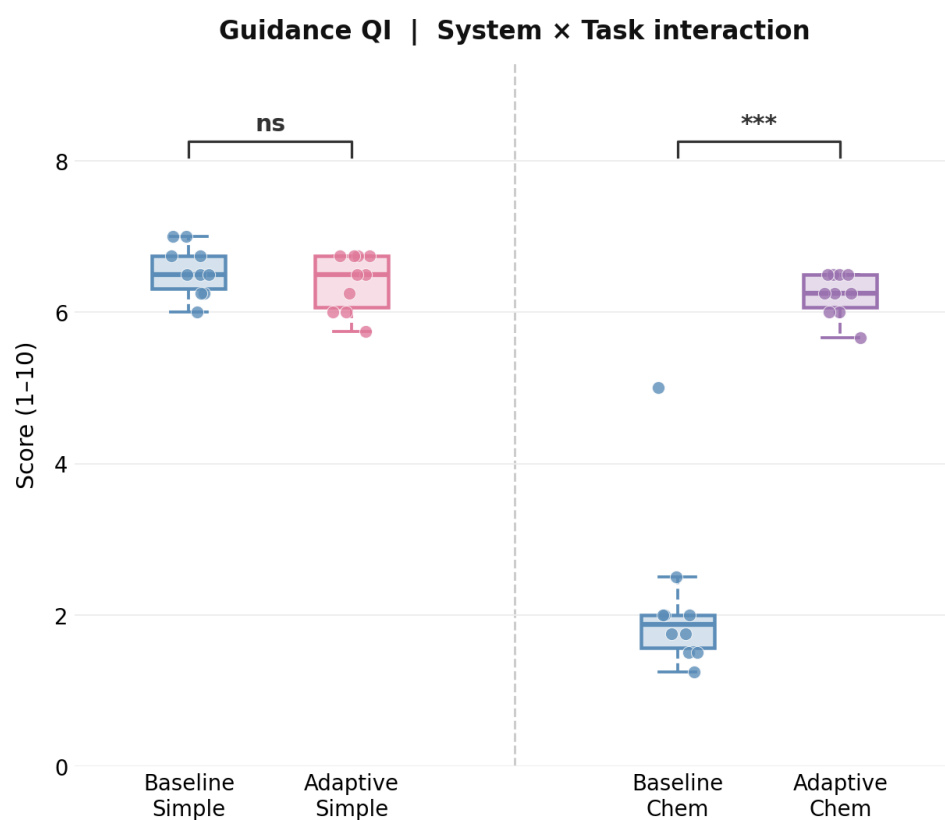


Figure 5.10: Guidance Quality Index scores across baseline and adaptive guidance conditions for simple and complex tasks.

Chapter 6

Discussion

6.1 Interpretation of Key Findings

The results demonstrated that combining semantic procedural tracking, multimodal behavioral sensing, and adaptive LLM-based guidance can improve procedural support in mixed reality environments. The proposed framework achieved reliable semantic action recognition and real-time procedural reasoning while maintaining low-latency interaction during runtime execution.

The findings additionally supported the central methodological assumption of the thesis: observable procedural difficulty can be inferred from multimodal behavioral interaction dynamics without directly estimating latent cognitive or affective states. The procedural difficulty signal consistently increased around procedural deviations and unstable interaction periods.

Behavioral analysis further suggested that adaptive guidance was particularly beneficial in the complex task condition, where procedural dependencies and recovery situations were more demanding. Compared with the non-adaptive baseline, the adaptive framework reduced procedural errors, shortened sustained struggling periods, and improved recovery behavior following deviations.

The evaluation also highlighted the importance of runtime procedural grounding for LLM-based assistance. Prompt configurations incorporating localized procedural context, recent interaction history, and procedural difficulty information produced substantially more relevant and contextually appropriate guidance than prompts relying only on static protocol descriptions.

6.2 Limitations and Future Work

Several limitations should be considered. First, the procedural difficulty inference model relied on proxy behavioral labels derived from HoloAssist annotations rather than direct cognitive-state measurements. Consequently, the inference output should be interpreted as an operational behavioral indicator rather than a validated cognitive-load estimate.

Second, the semantic interaction pipeline relied on task-specific procedural heuristics and a predefined set of supported semantic actions. Consequently, the proposed framework was primarily designed for structured tabletop procedural tasks with relatively constrained interaction spaces rather than unconstrained open-world environments.

Third, the present evaluation examined the adaptive guidance system as an integrated framework rather than as a set of isolated components. The adaptive condition combined several mechanisms, including context-aware instruction adaptation, voice-based guidance delivery, visual object cues, and runtime intervention timing. Therefore, the results should be interpreted as system-level evidence for the complete adaptive framework, rather than as causal evidence for the independent contribution of each individual mechanism. Controlled component-level ablation studies are needed in future work to isolate the relative effects of instruction adaptation, visual cues, voice-based guidance, and intervention timing.

Fourth, the adaptive guidance module relied on relatively small locally hosted language models to maintain deployment feasibility and low-latency inference. Although this improved runtime practicality, smaller models occasionally produced simplified or incomplete responses.

Future work should investigate stronger multimodal grounding methods, domain-specific fine-tuning using deployment data collected directly from MR procedural environments, and larger multimodal foundation models capable of jointly reasoning about language, spatial context, and procedural state. Long-term evaluations involving larger participant groups and more diverse procedural domains would additionally strengthen generalizability.

Chapter 7

Conclusion

This thesis presented a multimodal adaptive mixed reality guidance framework for procedural tasks involving real-world object interaction. The proposed system combined semantic procedural tracking, multimodal behavioral sensing, temporal procedural difficulty inference, and adaptive LLM-based guidance generation within a unified real-time mixed reality architecture deployed on Microsoft HoloLens 2.

The evaluation demonstrated that the proposed framework could reliably perform real-time semantic procedural tracking and adaptive guidance generation during procedural task execution. The results further showed that adaptive guidance reduced procedural errors, lowered sustained struggling periods, and improved recovery behavior compared with a non-adaptive baseline system, particularly during more complex chemistry-oriented tasks.

More broadly, this work demonstrated the feasibility of closed-loop adaptive procedural assistance in mixed reality environments using only non-invasive multimodal interaction signals available directly on commodity MR hardware. The proposed framework contributes toward future MR systems capable of interpreting user behavior, reasoning about procedural context, and dynamically adapting guidance during real-world task execution.

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Appendix

Prompt Templates

Model A — Full Protocol Baseline

You are a real-time lab instructor assistant in a Mixed Reality experiment.

TASK:

- Experiment: {protocol_name}
- Steps in this experiment:
{all_steps}

STUDENT QUESTION:

Student: {question}

INSTRUCTIONS:

- Answer based only on the protocol steps provided above.
- Be clear, concise, and actionable.
- Maximum 2 sentences.

Figure 1: Prompt template A.

Model B — Selective Runtime Context

You are a real-time lab instructor assistant in a Mixed Reality experiment.

TASK:

- Experiment: {protocol_name}
- Current step: {current_step}

RUNTIME STATE:

- Last action: {last_action}
- Last outcome: {last_outcome}
- Time on current step: {time_on_step} seconds
- Repeated attempts on current step: {repeat_count}
- Errors on current step: {error_count}
- Available next step(s): {next_steps}

STUDENT QUESTION:

Student: {question}

INSTRUCTIONS:

- Focus on the current step only.
- Available next step is for context; mention it only if explicitly asked.
- If the question cannot be answered from the current step, say only what the student should do now.
- Do not mention unrelated protocol steps.
- Be clear, concise, actionable. Maximum 2 sentences.

Figure 2: Prompt template B.

Model C — Full Adaptive Context

You are a real-time lab instructor assistant in a Mixed Reality experiment.

TASK:

- Experiment: {protocol_name}
- Current step: {current_step}

RUNTIME STATE:

- Last action: {last_action}
- Last outcome: {last_outcome}
- Time on current step: {time_on_step} seconds
- Repeated attempts on current step: {repeat_count}
- Errors on current step: {error_count}
- Available next step(s): {next_steps}

CURRENT STEP BREAKDOWN (use only if student asks for detail):

{current_sub_steps}

PREVIOUS EXCHANGE (use only if relevant to current question):

Student: {prev_question}

Instructor: {prev_answer}

STUDENT STATE:

- Procedural difficulty: {procedural_difficulty}

RESPONSE STYLE:

- Tone: {tone} [calm | direct | encouraging]
- Detail level: {detail} [very_short | concise | normal]
- Mode: {mode} [action_only | action_plus_short_reason]

STUDENT QUESTION:

Student: {question}

INSTRUCTIONS:

- Focus on the current step only.
- Do not mention unrelated protocol steps.
- Apply the response style above strictly.
- Maximum 2 sentences.

Figure 3: Prompt template C.

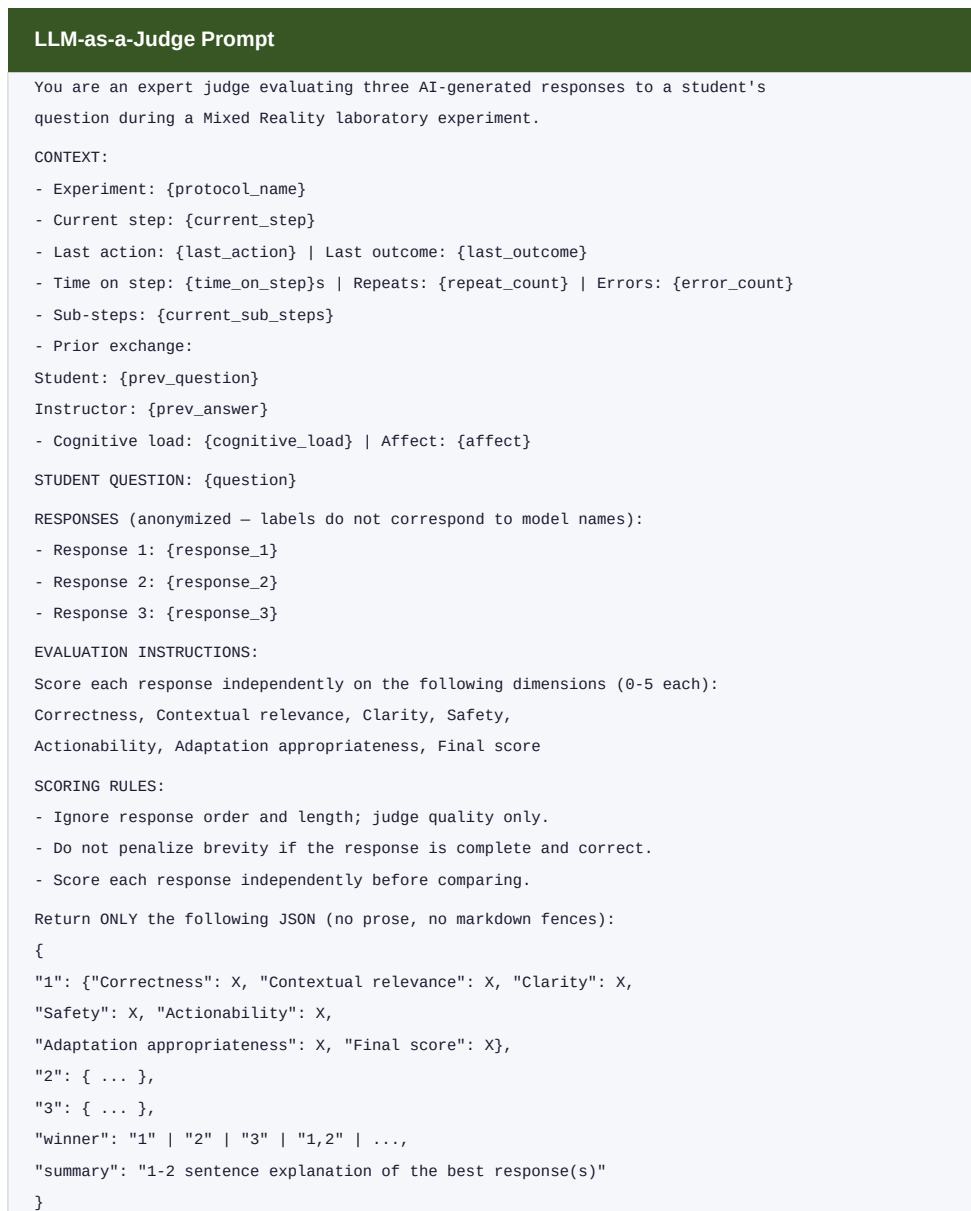


Figure 4: Judge prompt template.

User Study Materials

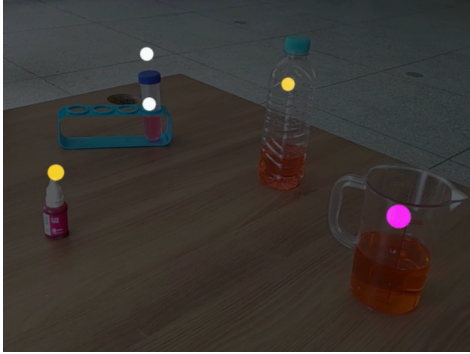


Figure 5: Demonstration of mixed reality visual indicators. Pink markers indicate that gaze is directed toward the object, and yellow markers indicate visual hints presented by the guidance system.

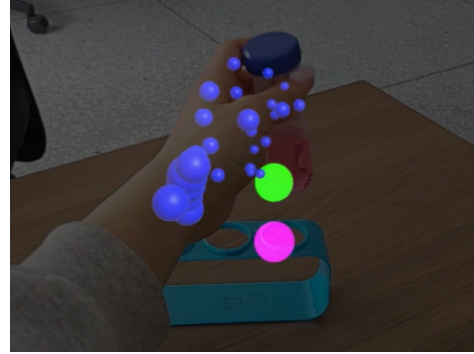


Figure 6: Demonstration of hand-object interaction indicators during runtime interaction. Green markers indicate detected hand-object interaction.

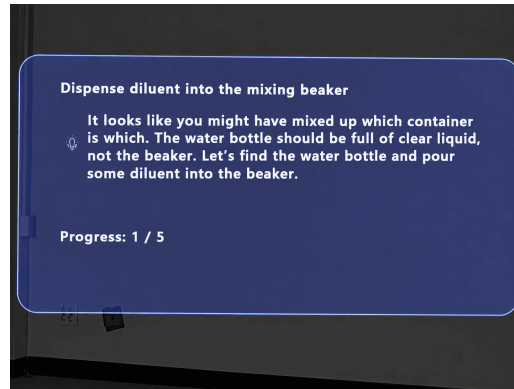


Figure 7: Runtime mixed reality guidance panel for presenting contextual procedural information and adaptive assistance to the user.

혼합현실에서의 절차적 작업 지원을 위한 다중모달 적응형 안내 시스템의 설계 및 구현

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요약

혼합현실(Mixed Reality, MR) 기반 안내 시스템은 주로 정적인 지시 방식에 의존하고 있어 실제 환경에서 사용자마다 나타나는 인지적·행동적 차이를 충분히 반영하지 못한다. 본 논문에서는 혼합현실 환경에서의 절차 기반 실험 교육을 지원하기 위한 다중모달 적응형 안내 시스템을 제안하며, 해당 시스템은 Microsoft HoloLens 2 플랫폼 상에서 구현되었다. 제안된 시스템은 구조화된 작업 표현, 실시간 의미 기반 단계 추적, 다중모달 사용자 상태 추정, 그리고 대규모 언어모델(LLM) 기반 적응형 안내를 하나의 통합된 폐쇄 루프(closed-loop) 구조로 결합한다. 작업 진행 상태는 자연어 프로토콜로부터 대규모 언어모델을 이용해 생성된 방향성 비순환 그래프(Directed Acyclic Graph, DAG)로 표현되며, 이를 통해 작업 절차를 자동적이고 유연하게 구조화한다. 또한 시선, 손, 머리 움직임으로부터 획득한 센서 데이터를 활용하는 다중모달 행동 인식 파이프라인을 통해 작업 단계를 실시간으로 추적한다. 이러한 다중모달 신호는 사용자의 절차 수행 난이도 상태를 추정하는 데 활용되며, 이를 기반으로 안내 시스템은 피드백의 세부 수준, 표현 방식 및 내용을 동적으로 조정한다. 안내 생성 모듈에 대한 평가 결과, 실시간 사용자 상태 정보와 작업 맥락 정보를 함께 활용할 경우 작업 맥락 정보만을 사용하는 기준 방식에 비해 응답의 관련성이 향상되고 환각(hallucination) 현상이 감소함을 확인하였다. 또한 실험 결과는 높은 단계 추적 성능과 낮은 시스템 지연 시간을 보여주었으며, 이를 통해 몰입형 실험 환경에서의 실시간 적용 가능성을 입증하였다. 종합적으로 본 연구는 사용자 상태를 고려한 적응형 MR 안내를 위한 실용적인 프레임워크를 제시하며, 획일적인 지시 제공 방식에서 벗어나 개인화된 학습 지원으로 발전하기 위한 기반을 마련한다.